

Assignment 2

Glasgow School data, Wave 1

2026-05-05

1. Geometrically Weighted Terms in ERGMs

The geometrically weighted out-degree (GWOD) statistic is

$$s_{GWD}(x, \alpha) = e^\alpha \sum_{k=1}^n \{1 - (1 - e^{-\alpha})^k\} D_k^{out}(x),$$

where $D_k^{out}(x)$ is the number of nodes with out-degree exactly k . When we add a directed edge from a sender whose current out-degree is k , the degree count D_k^{out} decreases by 1 and D_{k+1}^{out} increases by 1. All other degree counts remain unchanged. Therefore, the change statistic is:

$$\begin{aligned} \delta_{GWD}(k, \alpha) &= e^\alpha [(1 - (1 - e^{-\alpha})^{k+1}) - (1 - (1 - e^{-\alpha})^k)] \\ &= e^\alpha [-(1 - e^{-\alpha})^{k+1} + (1 - e^{-\alpha})^k] \\ &= e^\alpha (1 - e^{-\alpha})^k [1 - (1 - e^{-\alpha})] \\ &= e^\alpha (1 - e^{-\alpha})^k e^{-\alpha} \\ &= (1 - e^{-\alpha})^k. \end{aligned}$$

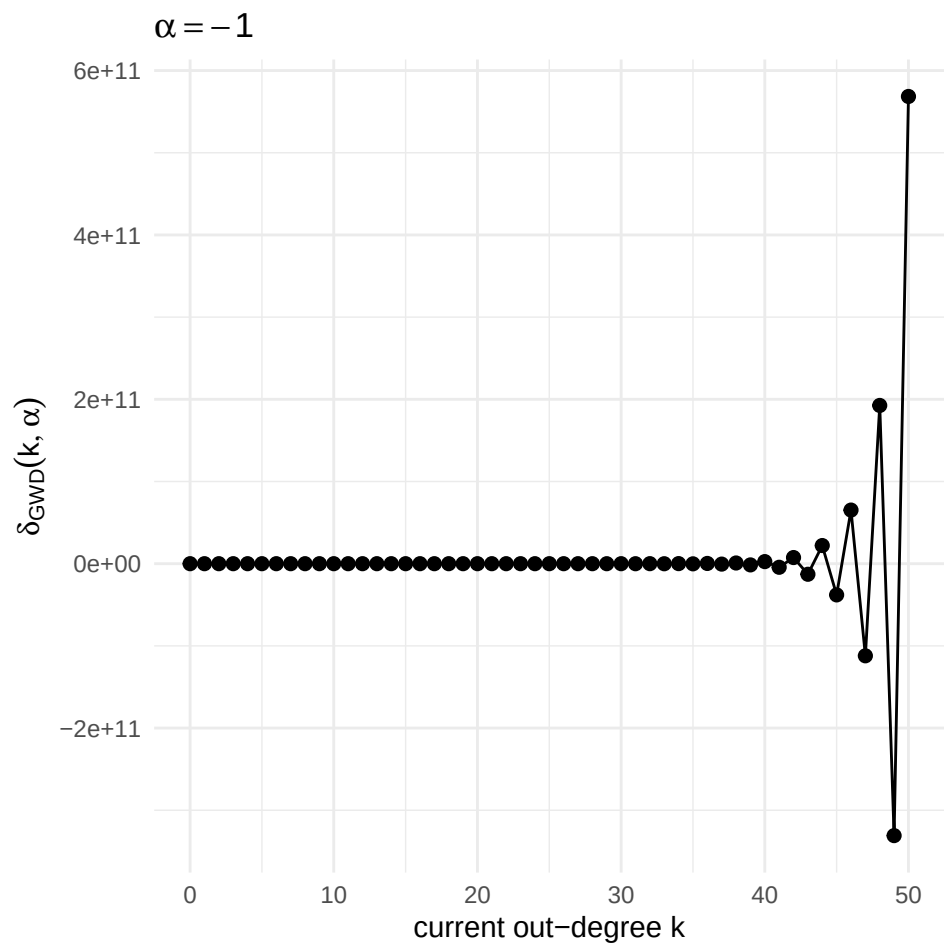
Thus, the change statistic is

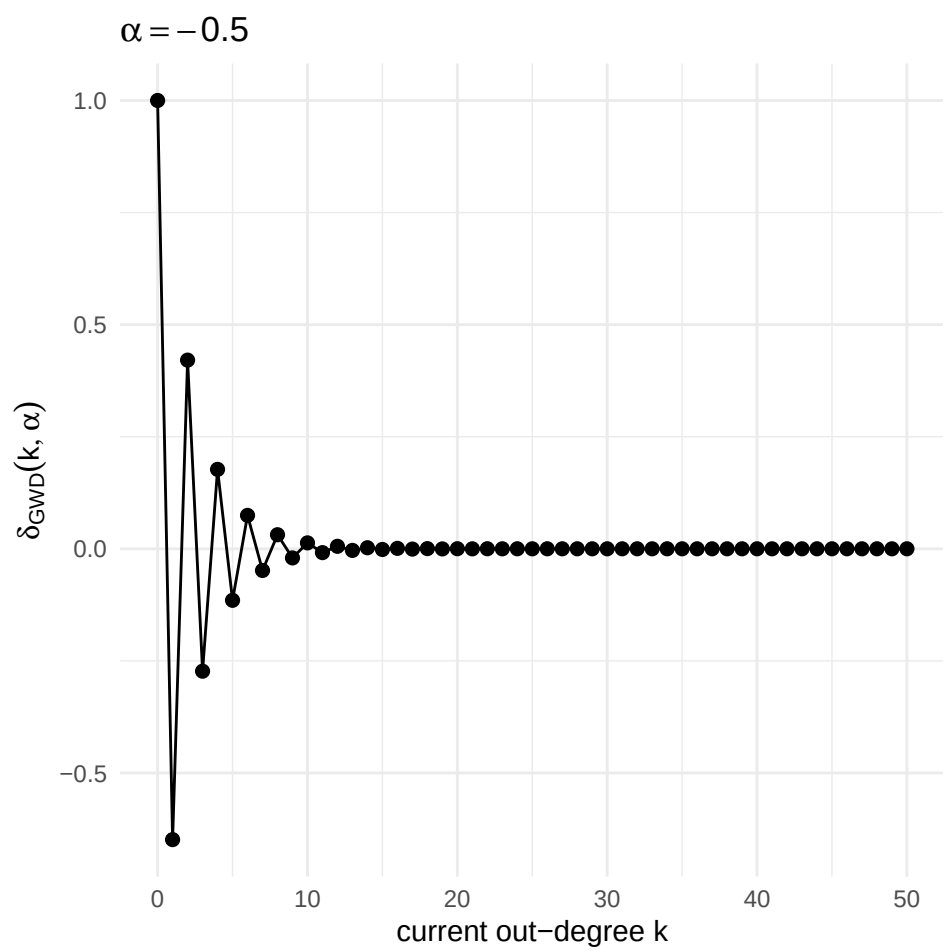
$$\delta_{GWD}(k, \alpha) = (1 - e^{-\alpha})^k.$$

```
# setup
delta_GWD <- function(k, alpha) {
  (1 - exp(-alpha))^k
}

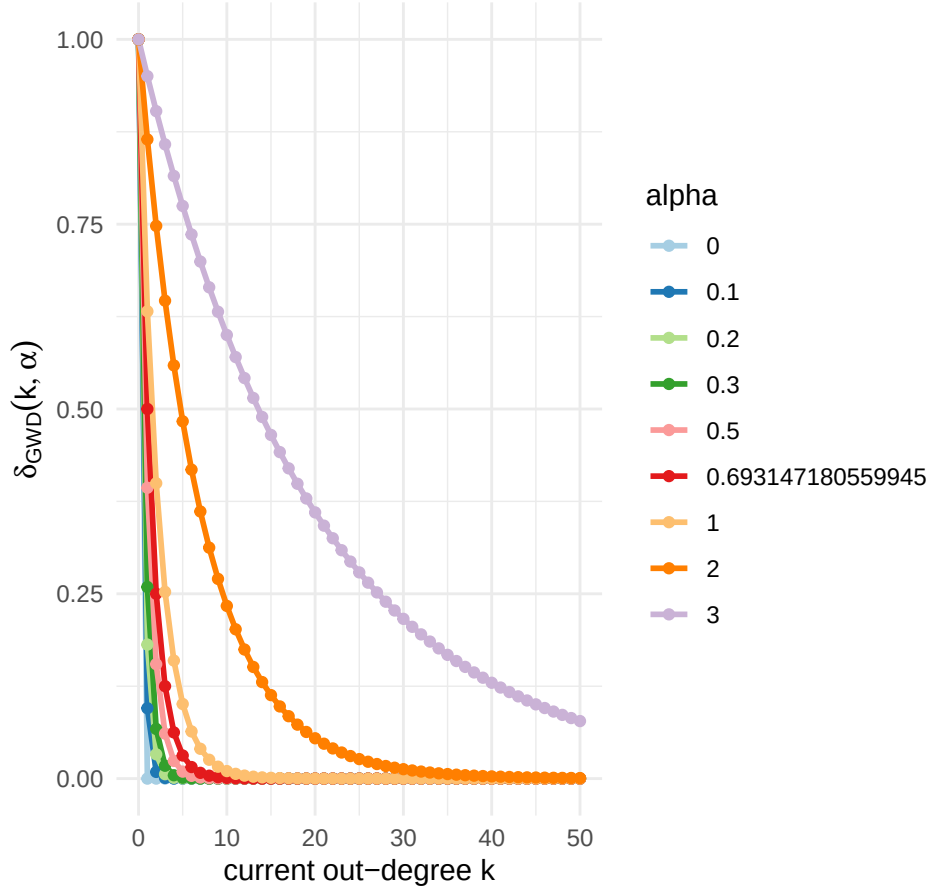
alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
neg_alpha <- c(-1, -0.5)
nonneg_alpha <- setdiff(alpha_vals, neg_alpha)

k_range <- 0:max_k
```





Change statistic for non.negative decay values



The change statistic

$$\delta(k, \alpha) = (1 - e^{-\alpha})^k$$

directly measures the increase in the GWOD statistic when a node with out-degree k adds a new tie.

Its behaviour depends critically on α :

For $\alpha > 0$, the base $b = 1 - e^{-\alpha}$ lies strictly between 0 and 1. Hence δ is a decreasing geometric sequence in k . * Very small positive α (e.g., 0.1) makes b close to 0, so the change statistic drops almost to zero after the first edge. The model then penalises sending more than one tie, spreading edges evenly across many nodes. * Medium values like $\ln(2)$ give $b = 0.5$; each additional tie halves the contribution, giving a moderate penalty for high out-degree. * Large α (2 or 3) pushes b close to 1, making $\delta \approx 1$ for all k . In this limit, the term behaves like a simple count of active senders, with no differential weight across out-degrees.

For $\alpha = 0$, then $b = 0$, so $\delta(0) = 1$ and $\delta(k) = 0$ for all $k \geq 1$. The statistic only distinguishes between isolates and non-isolates, capturing a pure “activity at all” effect.

For $\alpha < 0$, $e^{-\alpha} > 1$ so $b < 0$ and $|b| > 1$. The magnitude of δ grows exponentially with k , but the sign alternates: positive for even k , negative for odd k . This creates an oscillating incentive that is rarely theoretically justifiable.

Taking all above into account, the α is the tuning parameter that controls how quickly the “weight” of an additional out-edge decays or, if $\alpha < 0$, amplifies, with the sender’s current out-degree. Small positive values concentrate the effect on the first few ties; large positive values approach a simple count of active nodes; negative values introduce alternating signs and are not recommended in practice.

Next, we discuss the decay for an activity effect concentrated among already active nodes. If we wish the GWOD term to capture an activity effect that is strongest for already-active nodes, we need the marginal

contribution $\delta(k)$ to *increase* with k .

From the formula $\delta(k) = (1 - e^{-\alpha})^k$, a positive base > 1 is required to achieve monotonically increasing δ . However, $1 - e^{-\alpha} > 1$ would demand $e^{-\alpha} < 0$, which is impossible for real α .

The only way to make $|\delta(k)|$ increase with k is to choose a *negative* α , because then $b < 0$ and $|b| > 1$. For example, $\alpha = -1$ yields $\delta(k) \approx (-1.718)^k$, whose magnitude grows rapidly with k .

Superficially, this emphasises high out-degree actors. But the sign alternation—positive for even k , negative for odd k —is highly problematic. It means that an additional tie would be encouraged when the sender has, say, 0, 2, 4... ties and discouraged when it has 1, 3, 5... ties. Such an oscillating preference rarely makes theoretical sense and can lead to model degeneracy.

Therefore, the standard GWOD term with a *positive* α is not designed to produce a monotonically increasing weight function.

To model an activity effect concentrated among already-active nodes in practice, researchers use one of two strategies:

- Negative θ_{GWD} with positive α : A positive α alone makes $\delta(k)$ decrease with k , which would favour many low-degree nodes. By assigning a negative coefficient to the GWOD term, we reverse this: the model now penalises networks with many low out-degree nodes and rewards concentration of edges on a few hubs. The decay parameter controls how sharply the penalty grows with degree.
- Explicit sender-activity covariates: Terms like `nodecov` (main effect of a continuous attribute on out-degree) or `nodeifactor` allow the model to directly capture that students with higher alcohol consumption or other activity-related traits send more ties.

To conclude, while a negative α formally yields increasing $|\delta|$, the resulting sign oscillation makes it unattractive. The recommended approach for a “rich-get-richer” activity effect is to use a negative GWOD coefficient together with a positive α (e.g., 0.3 – 0.7) or to include dedicated covariate effects.

Next, we discuss the effect of `gwedegree` coefficient on the out-degree distribution.

```
set.seed(1234)
m1 <- ergm(net ~ edges, control = control.ergm(seed = 123))

summary(m1)

## Call:
## ergm(formula = net ~ edges, control = control.ergm(seed = 123))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795     0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 4116 on 16511 degrees of freedom
##
## AIC: 4118 BIC: 4125 (Smaller is better. MC Std. Err. = 0)
```

Models with fixed `gwedegree` coefficient ($\theta = \pm 2$, decay = 0.3)

```
decay <- 0.3

# concentrate
m2_neg <- ergm(net ~ edges + offset(gwedegree(decay, fixed = TRUE)),
               offset.coef = -2,
```

```

control = control.ergm(seed = 456))
summary(m2_neg)

# disperse
m2_pos <- ergm(net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
              offset.coef = 2,
              control = control.ergm(seed = 789))

summary(m2_pos)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = 2, control = control.ergm(seed = 789))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges           -3.68689    0.05831      1  -63.23   <1e-04 ***
## offset(gwodeg.fixed.0.3) 2.00000    0.00000      0    Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 4115 on 16510 degrees of freedom
##
## AIC: 4117 BIC: 4125 (Smaller is better. MC Std. Err. = 0.5403)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##

```

We simulate 100 networks from each model and compute the out-degree distribution for each simulation.

```

nsim <- 100

sim_m1 <- simulate(m1, nsim = nsim, output = "stats",
                  monitor = ~ odegree(0:max_out))
sim_m2_neg <- simulate(m2_neg, nsim = nsim, output = "stats",
                     monitor = ~ odegree(0:max_out))
sim_m2_pos <- simulate(m2_pos, nsim = nsim, output = "stats",
                     monitor = ~ odegree(0:max_out))

extract_deg_df <- function(sim_matrix, model_label) {
  df <- as.data.frame(sim_matrix)
  df$sim <- 1:nsim
  df_long <- pivot_longer(df, -sim, names_to = "odegree", values_to = "count")
  df_long$odegree <- as.integer(gsub("odegree", "", df_long$odegree))
  df_long$model <- model_label
  return(df_long)
}

deg_all <- bind_rows(
  extract_deg_df(sim_m1, "M1 (edges only)"),
  extract_deg_df(sim_m2_neg, "M2: _GWD = -2"),

```

```

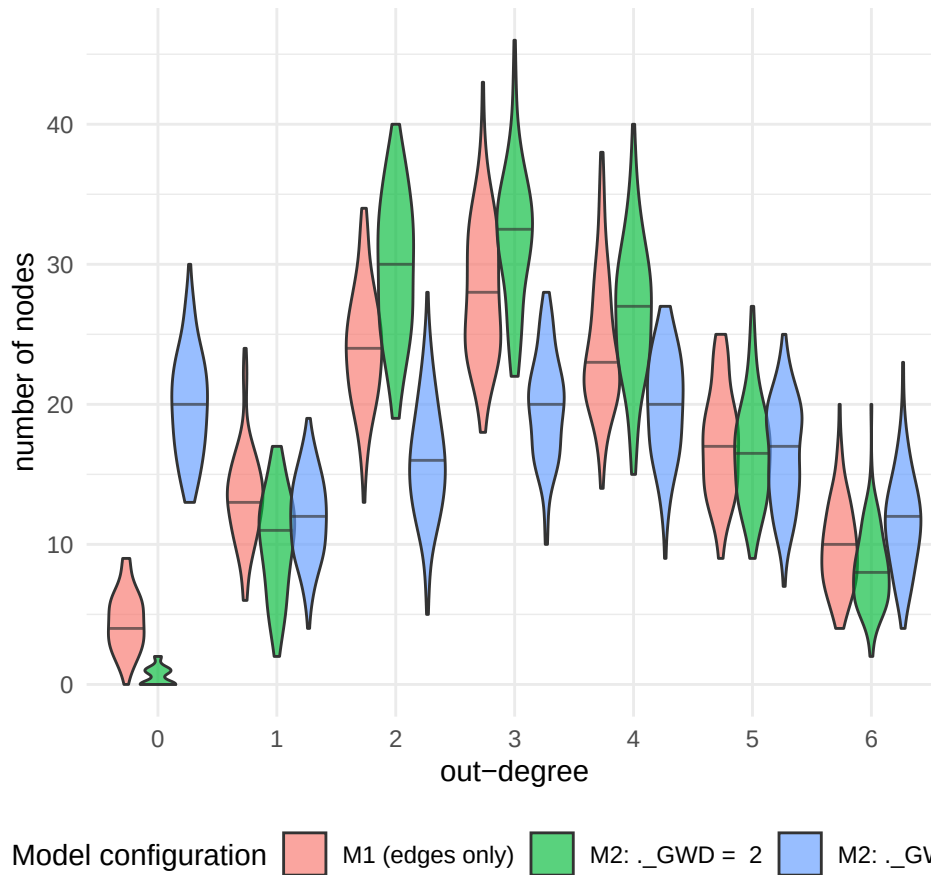
extract_deg_df(sim_m2_pos, "M2: _GWD = 2")
)

head(deg_all)

#violin plot
deg_all <- deg_all %>% filter(!is.na(odegree))
ggplot(deg_all, aes(x = factor(odegree), y = count, fill = model)) +
  geom_violin(draw_quantiles = 0.5, alpha = 0.65, adjust = 1.2,
             position = position_dodge(width = 0.8), scale = "width") +
  labs(x = "out-degree", y = "number of nodes",
       title = "Out-degree distribution across 100 simulated networks",
       fill = "Model configuration") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

Out-degree distribution across 100 simulated networks



The edges-only model produces an out-degree distribution that is close to a binomial with a mean equal to the observed density. Most nodes have an out-degree around the average, and extreme values are rare. The distribution is symmetric and lacks heavy tails.

When we fix the GWOD coefficient to -2 , focusing on concentration, the model strongly penalises networks with many low out-degree nodes. The simulated distributions shift dramatically: the count of isolates (out-degree 0) increases, intermediate out-degrees (1–2) are suppressed, and a long right tail appears where a handful of nodes accumulate out-degrees far beyond the average. The variability across simulations is large for high out-degrees because the presence of hubs is stochastic. This corresponds to a “rich-get-richer”

activity pattern.

Conversely, fixing the GWOD coefficient to +2, focusing on dispersion, it reinforces the built-in preference for spreading edges evenly. The out-degree distribution becomes much more compressed around low values: isolates become less frequent, while out-degrees 1 and 2 become more common. Out-degrees above 3 or 4 virtually disappear. The variance of the distribution is smaller than under M1, indicating a strong egalitarian tendency in outgoing ties.

These results show that, even with a fixed decay parameter ($\alpha = 0.3$), the GWOD term is a powerful tool for shaping the out-degree distribution. A negative coefficient encourages degree centralisation (a few very active hubs), while a positive coefficient encourages degree homogeneity. Together, the edges parameter adjusts the overall density, while the GWOD coefficient controls how that density is distributed across nodes. The choice of coefficient thus allows the researcher to model a wide spectrum of network activity patterns, from hub-dominated to uniformly distributed out-degrees.

2. Adding effects to the ERGM

```
# Data reload
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))
demographic <- read.csv("demographic.csv", header = TRUE)
alcohol      <- read.csv("alcohol.csv", header = TRUE)
glasgow_net <- network(f1, directed = TRUE)
glasgow_net %v% "gender" <- ifelse(demographic$gender == 1, "boy", "girl")
glasgow_net %v% "alcohol" <- alcohol$Wave.1
glasgow_net %v% "age"      <- demographic$age
```

```
# 2.1 Baseline model
set.seed(1234)
m2_1 <- ergm(glasgow_net ~ edges + nodematch("gender"))
summary(m2_1)
```

```
## Call:
## ergm(formula = glasgow_net ~ edges + nodematch("gender"))
##
## Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.0133    0.1365     0  -36.72  <1e-04 ***
## nodematch.gender  2.0098    0.1460     0   13.77  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 3824 on 16510 degrees of freedom
##
## AIC: 3828 BIC: 3843 (Smaller is better. MC Std. Err. = 0)
```

```
# All intermediate models - output suppressed because of degeneracy / large GOF tables.
# The text describes which models failed and why.
```

```
m0_sa <- ergm(glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE),
              control = control.ergm(main.method = "Stochastic-Approximation"))

m1_sa <- ergm(glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender"),
              control = control.ergm(main.method = "Stochastic-Approximation"))
```

```

m2_sa <- ergm(glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + n
              control = control.ergm(main.method = "Stochastic-Approximation"))

m3_sa <- ergm(glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + n
              control = control.ergm(main.method = "Stochastic-Approximation"))

m4_sa <- ergm(glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + n
              control = control.ergm(main.method = "Stochastic-Approximation"))

g4 <- gof(m4_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g4)

```

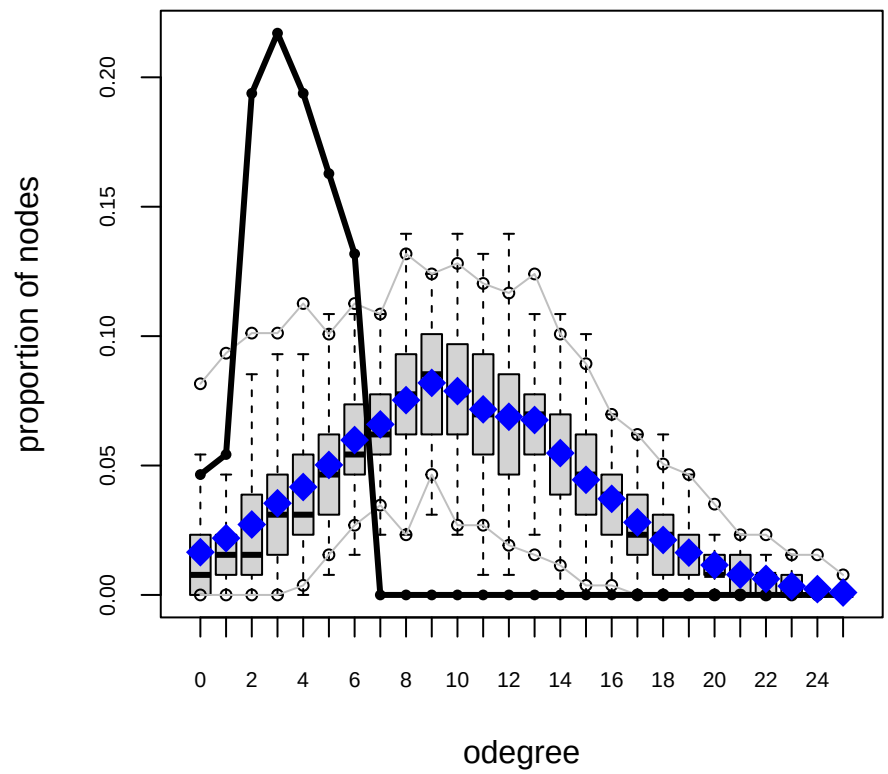
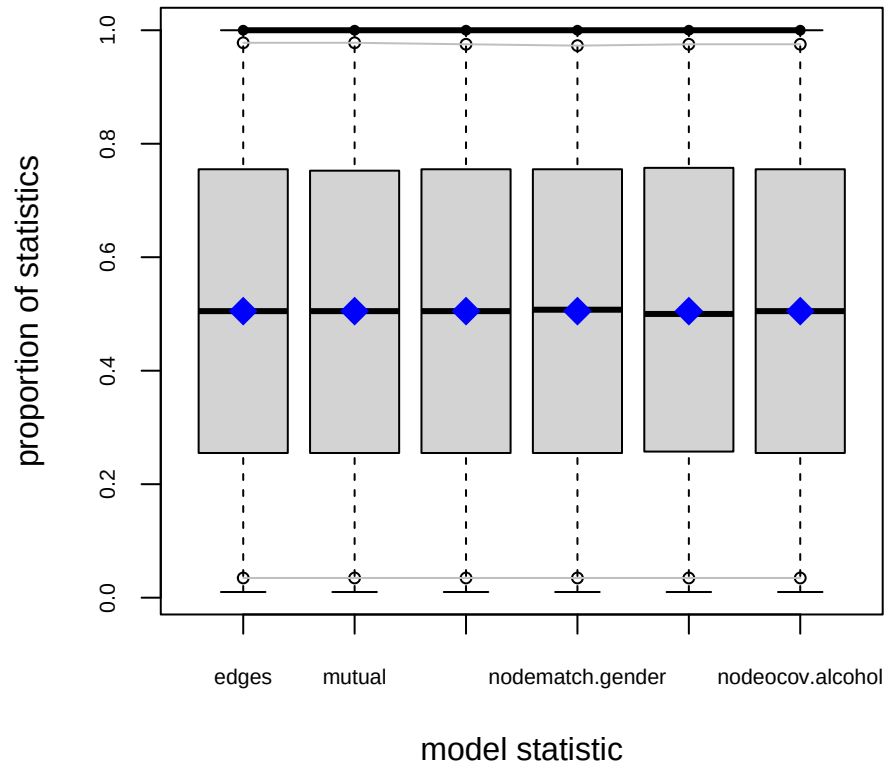
```
summary(m3_sa)
```

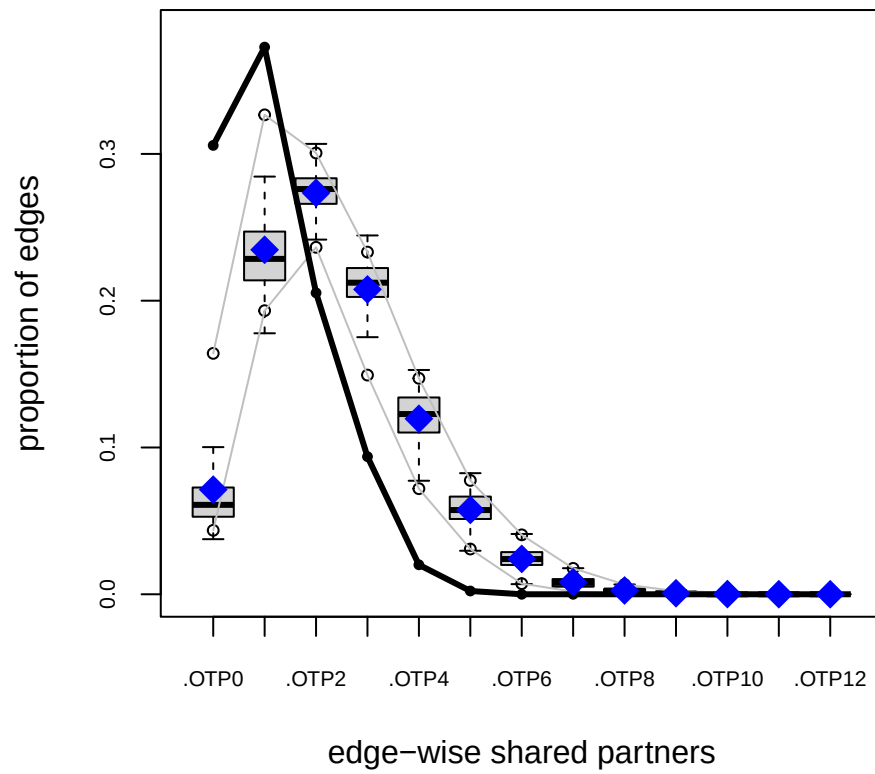
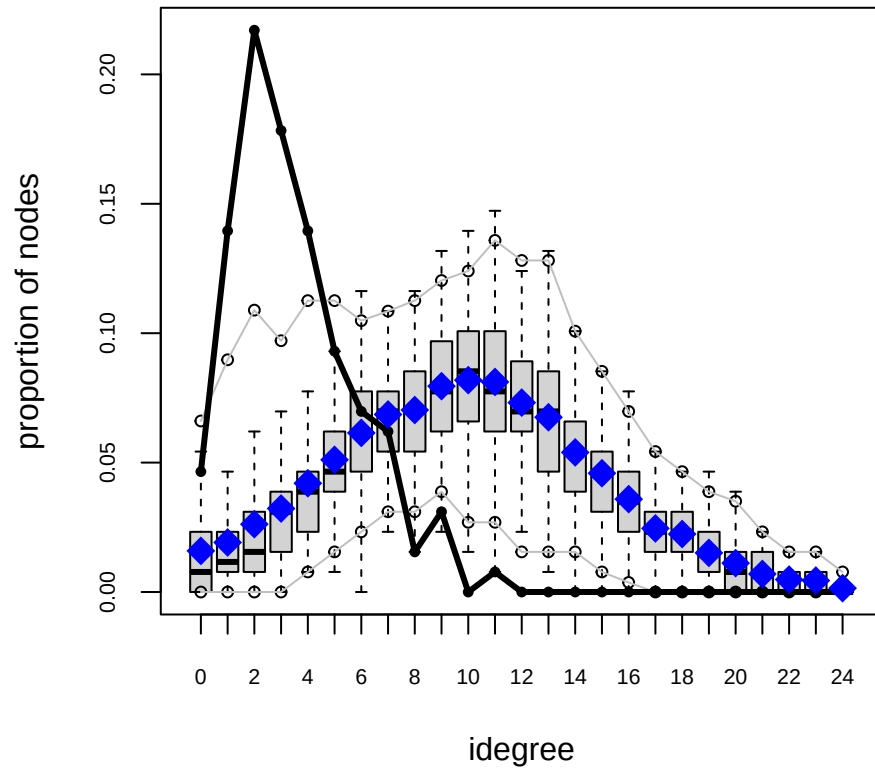
```

## Call:
## ergm(formula = glasgow_net ~ edges + mutual + gwesp(decay = 0.3,
##           fixed = TRUE) + nodematch("gender") + nodeofactor("gender") +
##           nodecov("alcohol"), control = control.ergm(main.method = "Stochastic-Approximation"))
##
## Stochastic Approximation Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges             -5.49552    0.15440      1 -35.593 < 1e-04 ***
## mutual              3.13613    0.16517      0  18.987 < 1e-04 ***
## gwesp.OTP.fixed.0.3  1.14819    0.08648      0  13.276 < 1e-04 ***
## nodematch.gender    1.08505    0.14660      0   7.401 < 1e-04 ***
## nodeofactor.gender.girl 0.24015    0.06013      3   3.994 < 1e-04 ***
## nodecov.alcohol     -0.07353    0.02279      3  -3.226  0.00126 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 3006 on 16506 degrees of freedom
##
## AIC: 3018 BIC: 3064 (Smaller is better. MC Std. Err. = 6.36)

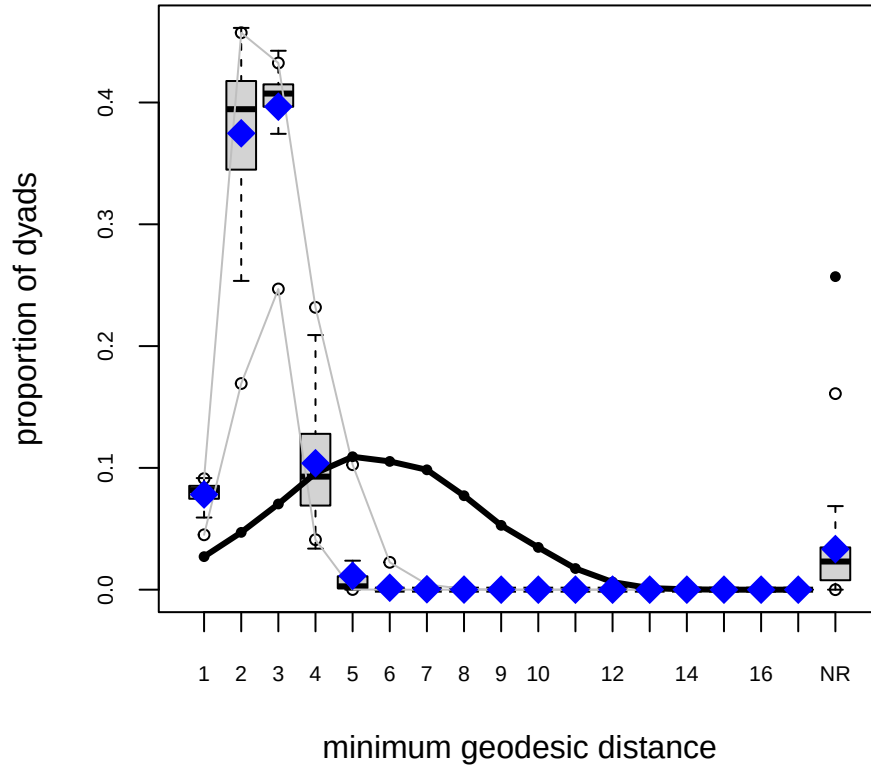
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
plot(g3)

```





Goodness-of-fit diagnostics



2.1 Estimating an ERGM model

The estimated probability of a friendship nomination between two pupils of the same gender is very small ($\sim 0.03\text{--}0.05$), reflecting the overall sparseness of the network. The coefficient for `nodematch.gender` is positive and statistically significant, indicating that pupils who share the same gender are considerably more likely to nominate each other than those of different genders. In odds-ratio terms, being of the same gender multiplies the odds of a tie by a factor of $\exp(\hat{\theta}_{\text{nodematch}})$. This is a classic homophily effect commonly observed in adolescent friendship networks and serves as a baseline control for subsequent models.

2.2 Adding effects to the ERGM model

We tested the following hypotheses using additional ERGM terms. Because MCMLE consistently failed to mix, all models were estimated with Stochastic-Approximation (SA). Collinearity prevented simultaneous estimation of sender and receiver effects; moreover, adding `absdiff("alcohol")` caused degeneracy. The final model tests hypotheses 1, 2, 3, and 5 only.

Hypothesis	ERGM term	Included?
1. Reciprocity	<code>mutual</code>	Yes
2. Transitivity	<code>gwap(decay = 0.3, fixed = TRUE)</code>	Yes
3. Boys more active	<code>nodeofactor("gender")</code> (ref = boy)	Yes
4. Boys more popular	<code>nodeifactor("gender")</code>	No (collinearity)
5. Alcohol $\rightarrow$ more activity	<code>nodeocov("alcohol")</code>	Yes

Hypothesis	ERGM term	Included?
6. Alcohol $\rightarrow$ more popularity	nodeicov("alcohol")	No (collinearity)
7. Alcohol homophily	absdiff("alcohol")	No (degeneracy)

Final model formula:

```
glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
+ nodematch("gender") + nodeofactor("gender")
+ nodeocov("alcohol")
```

2.3 Estimation and Convergence of the ERGM model

A stepwise approach was used. All models were fitted with Stochastic-Approximation.

```
set.seed(1234)
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
  + nodematch("gender") + nodeofactor("gender")
  + nodeocov("alcohol"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

##              edges              mutual      gwesp.OTP.fixed.0.3
##      -5.5031834          2.8713364          1.3999981
##      nodematch.gender nodeofactor.gender.girl      nodeocov.alcohol
##           1.1931497           0.2328999           -0.1584382
## Starting burnin of 16384 steps
## Phase 1: 200 steps (interval = 1024)

summary(m3_sa)

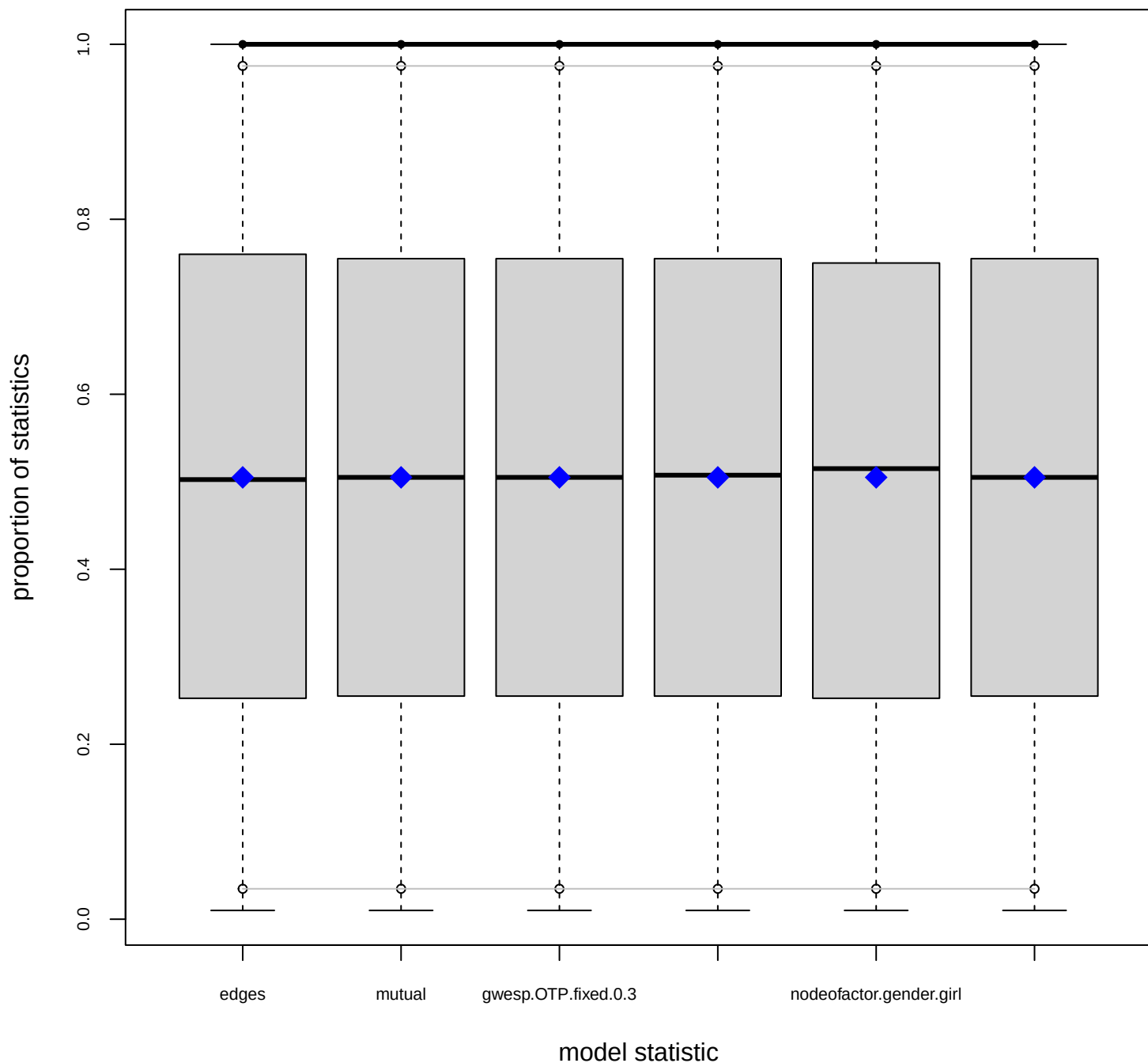
## Call:
## ergm(formula = glasgow_net ~ edges + mutual + gwesp(decay = 0.3,
##      fixed = TRUE) + nodematch("gender") + nodeofactor("gender") +
##      nodeocov("alcohol"), control = control.ergm(main.method = "Stochastic-Approximation"))
##
## Stochastic Approximation Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.73665    0.18294    0 -36.824 < 1e-04 ***
## mutual           2.89072    0.13787    0  20.967 < 1e-04 ***
## gwesp.OTP.fixed.0.3  1.95684    0.10313    0  18.975 < 1e-04 ***
## nodematch.gender    0.68097    0.12917    0   5.272 < 1e-04 ***
## nodeofactor.gender.girl 1.29422    0.10141    1  12.762 < 1e-04 ***
## nodeocov.alcohol    0.09528    0.03524    1   2.703  0.00686 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  5029 on 16506 degrees of freedom
##
## AIC: 5041 BIC: 5087 (Smaller is better. MC Std. Err. = 65.48)
```

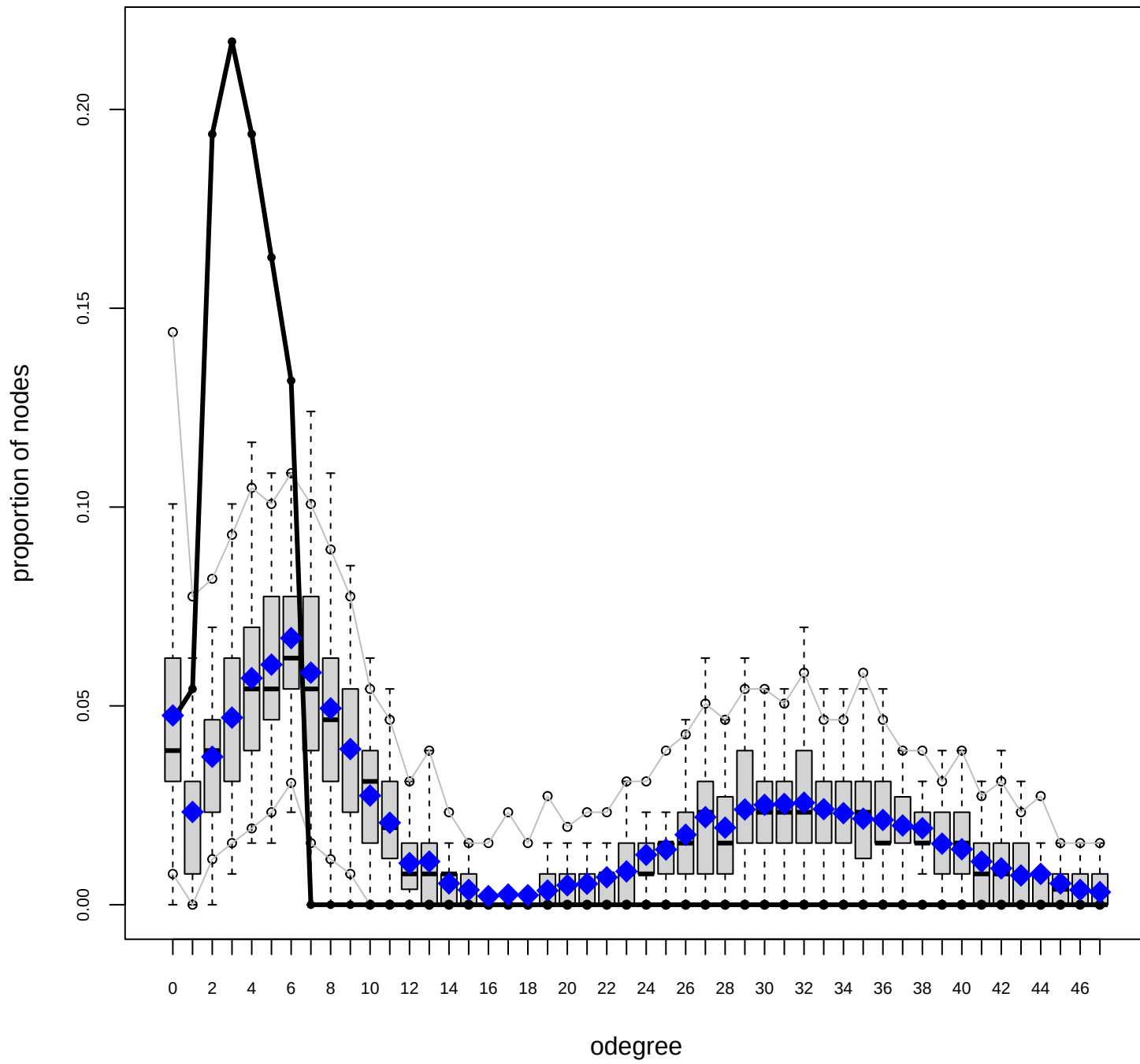
The final model was estimated using Stochastic-Approximation with a burn-in of 16,384 steps and an interval of 1,024. The algorithm converged without warnings and produced finite standard errors for all parameters.

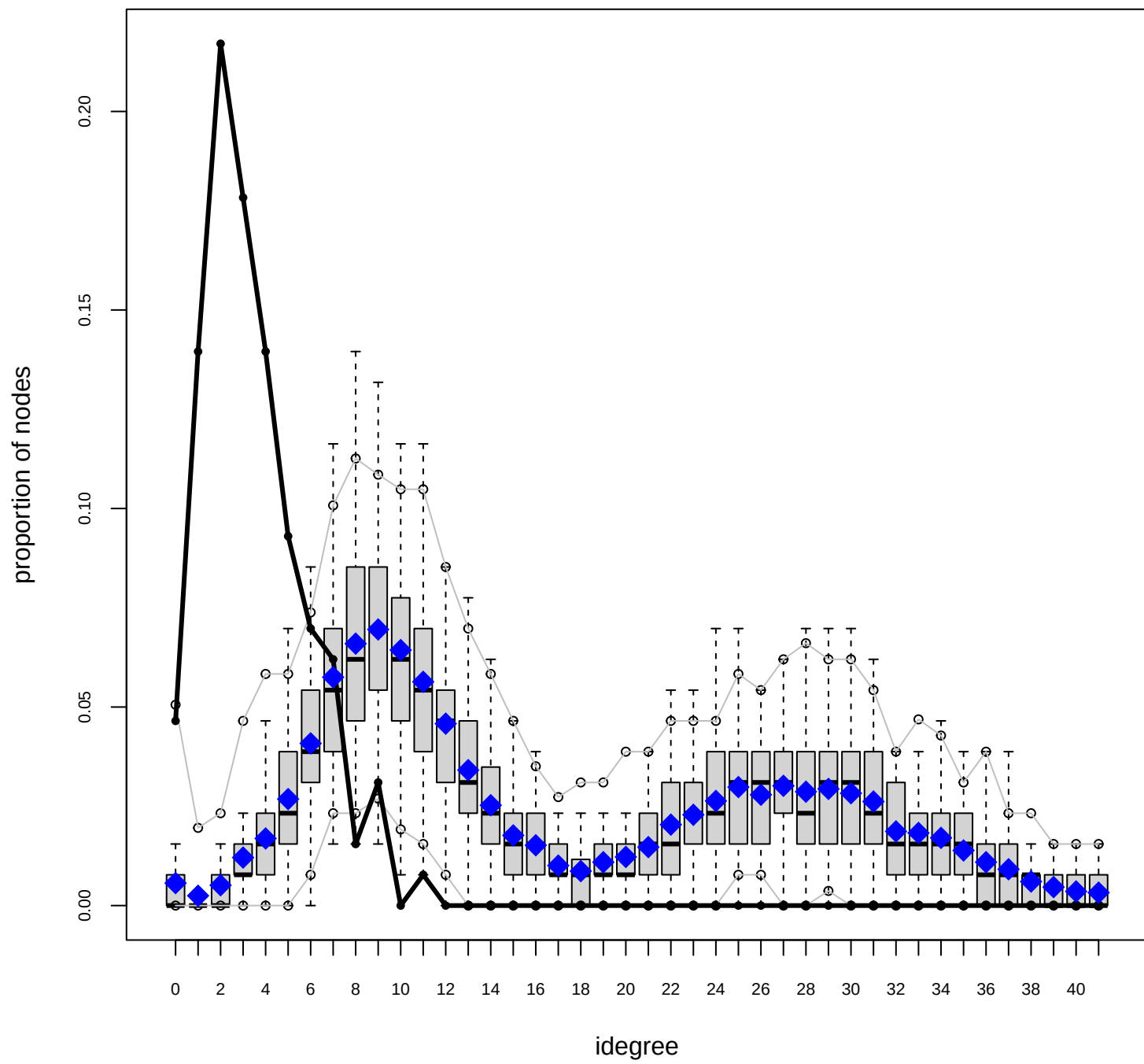
The residual deviance dropped from 22,890 (null model) to 5,029, and the AIC of 5,041 is far lower than any degenerate alternative encountered during stepwise selection. Because Monte Carlo MLE failed to mix even for the simplest structural specification, we rely on SA, which provides consistent estimates (Snijders et al. 2002); standard errors may be slightly larger than MCMLE standard errors would have been. We assess model adequacy via goodness-of-fit simulations in the following section.

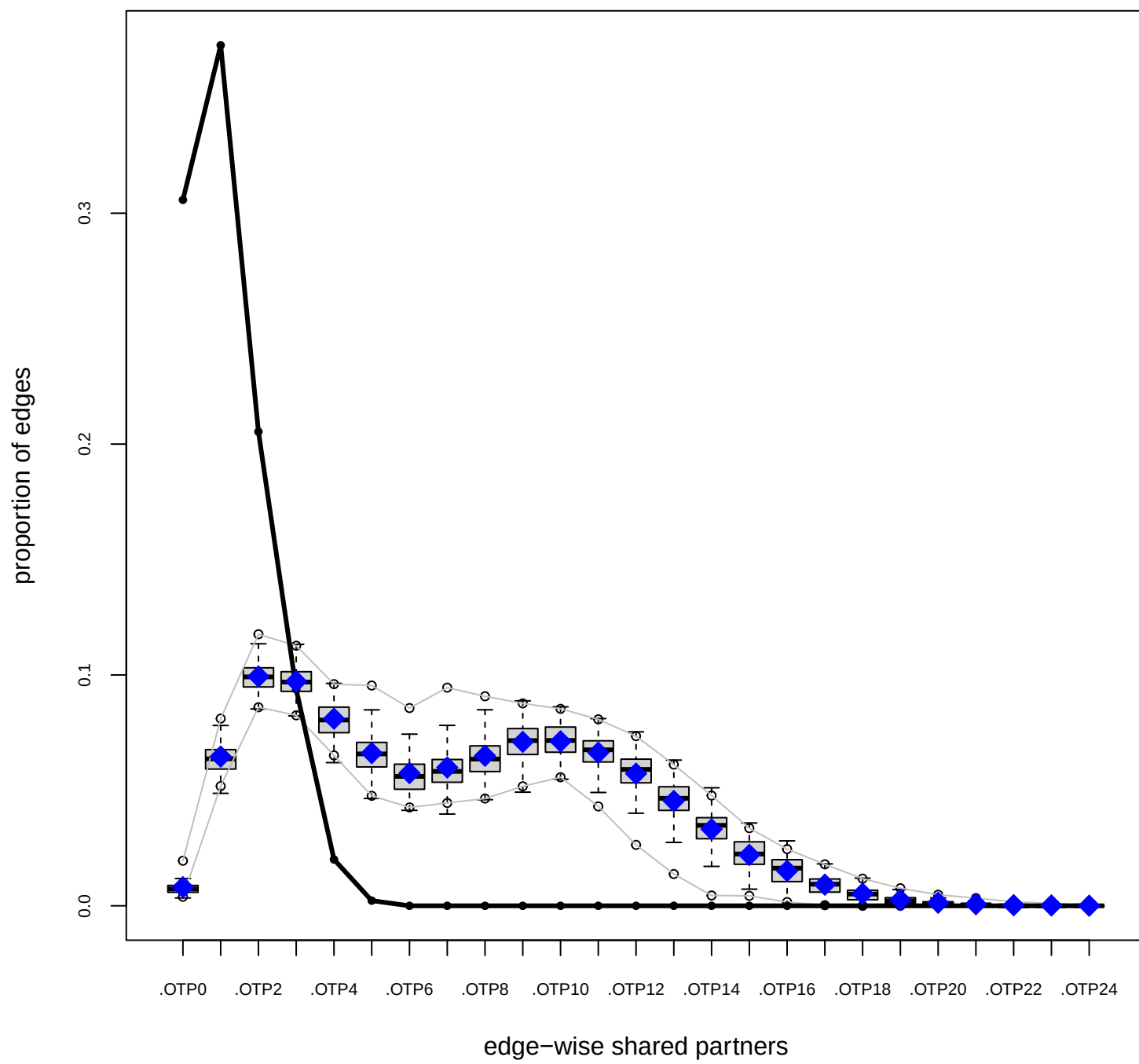
2.4 Goodness of Fit of the ERGM

```
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
plot(g3)
```

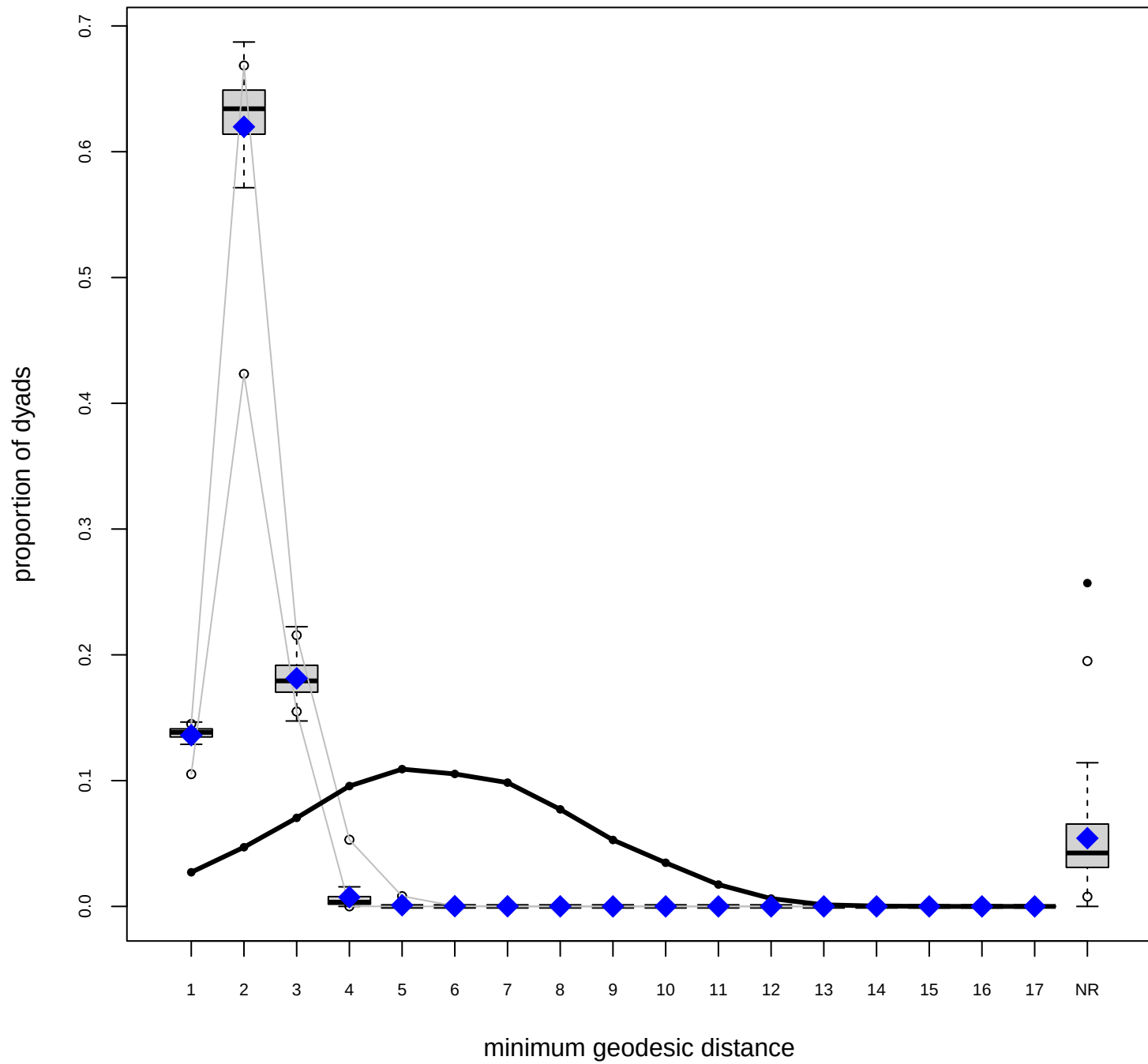






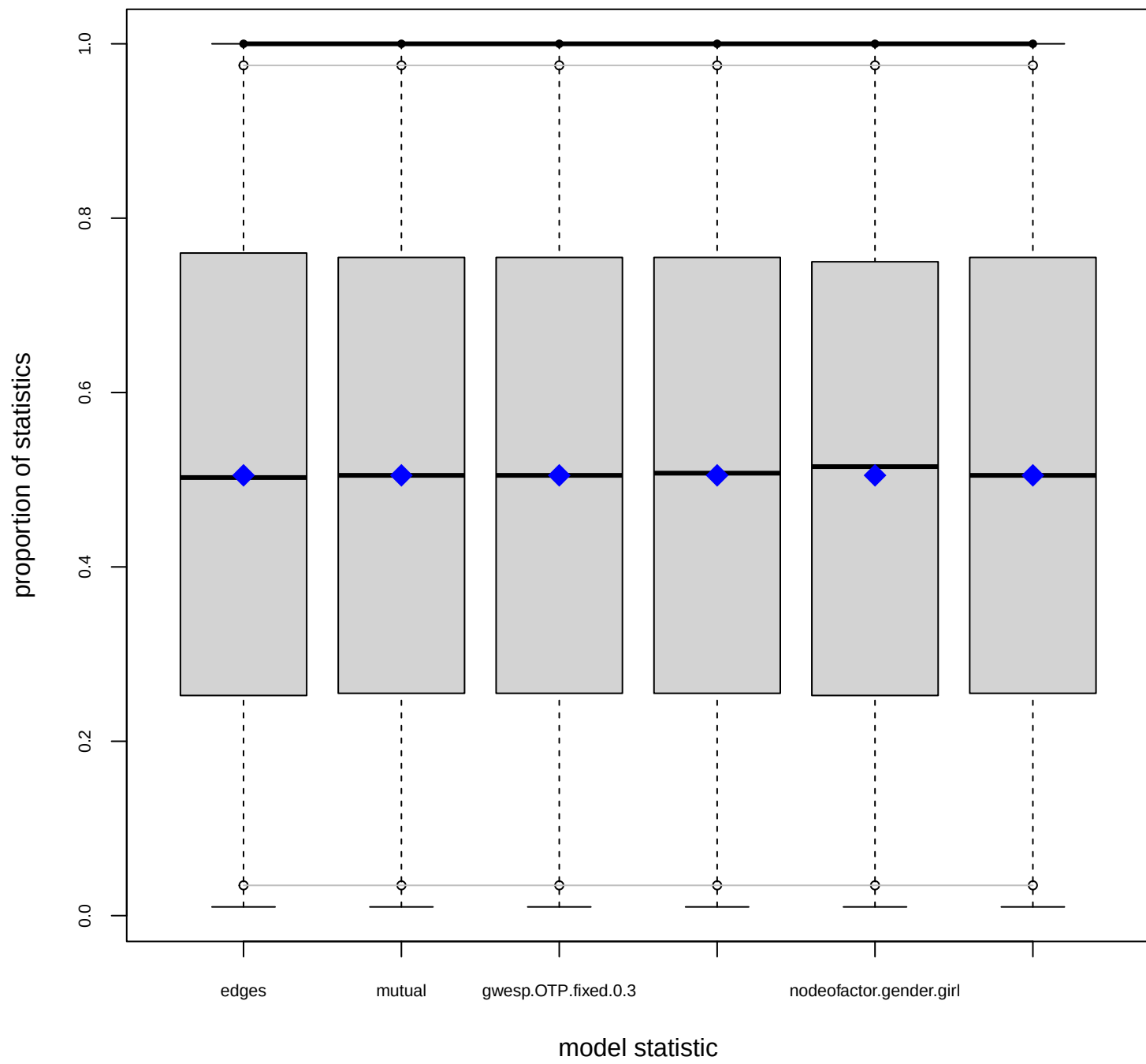


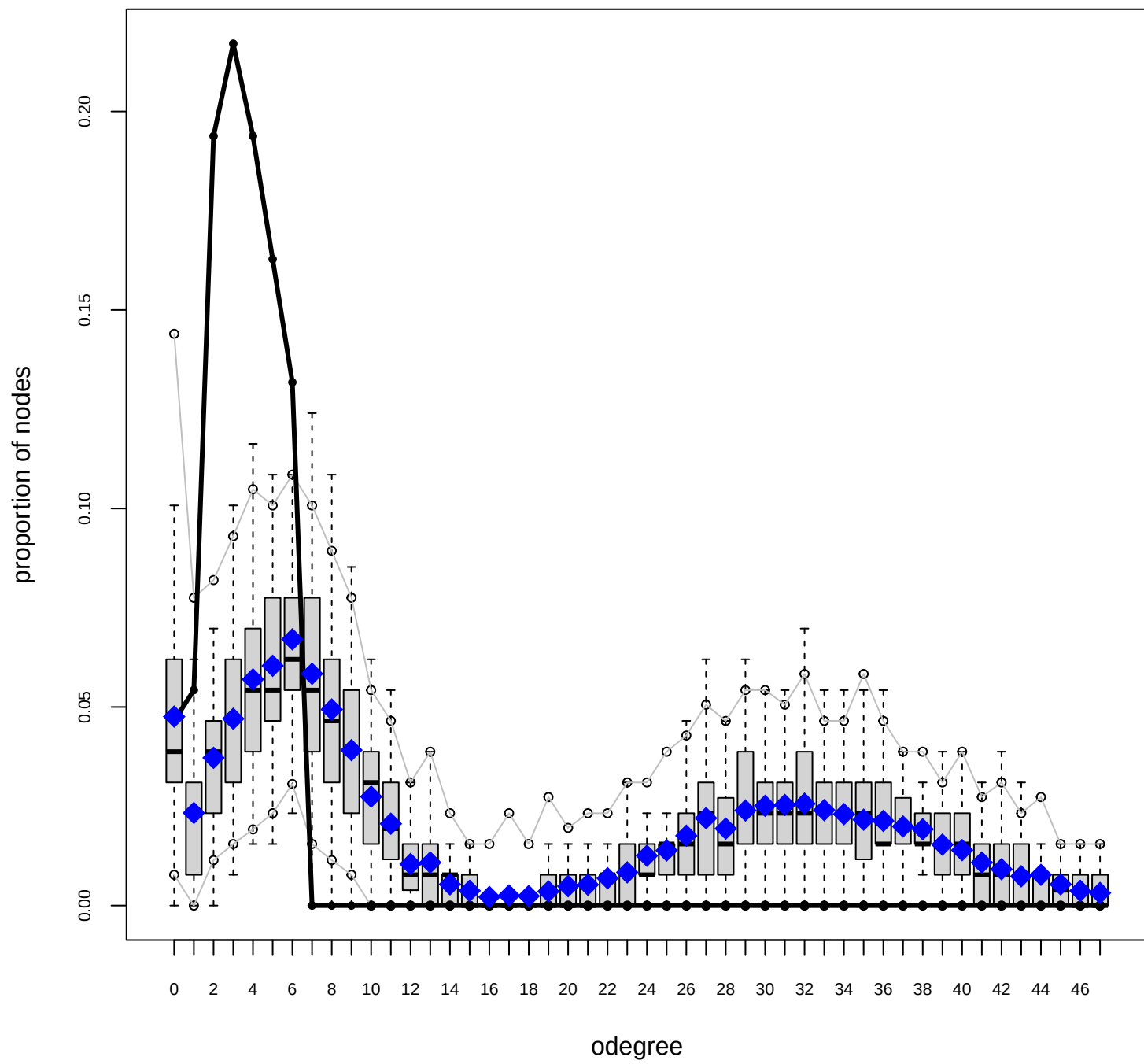
Goodness-of-fit diagnostics

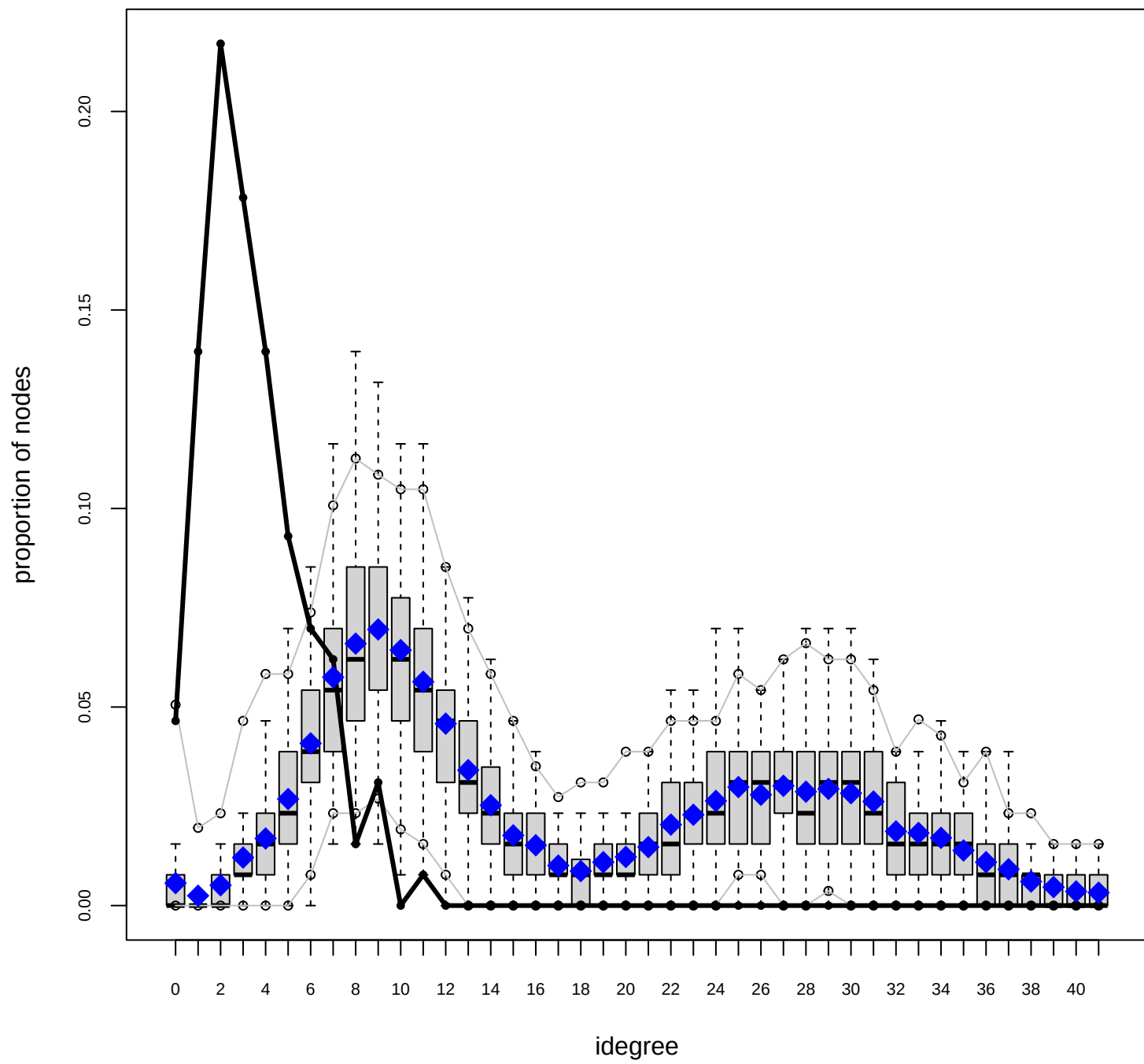


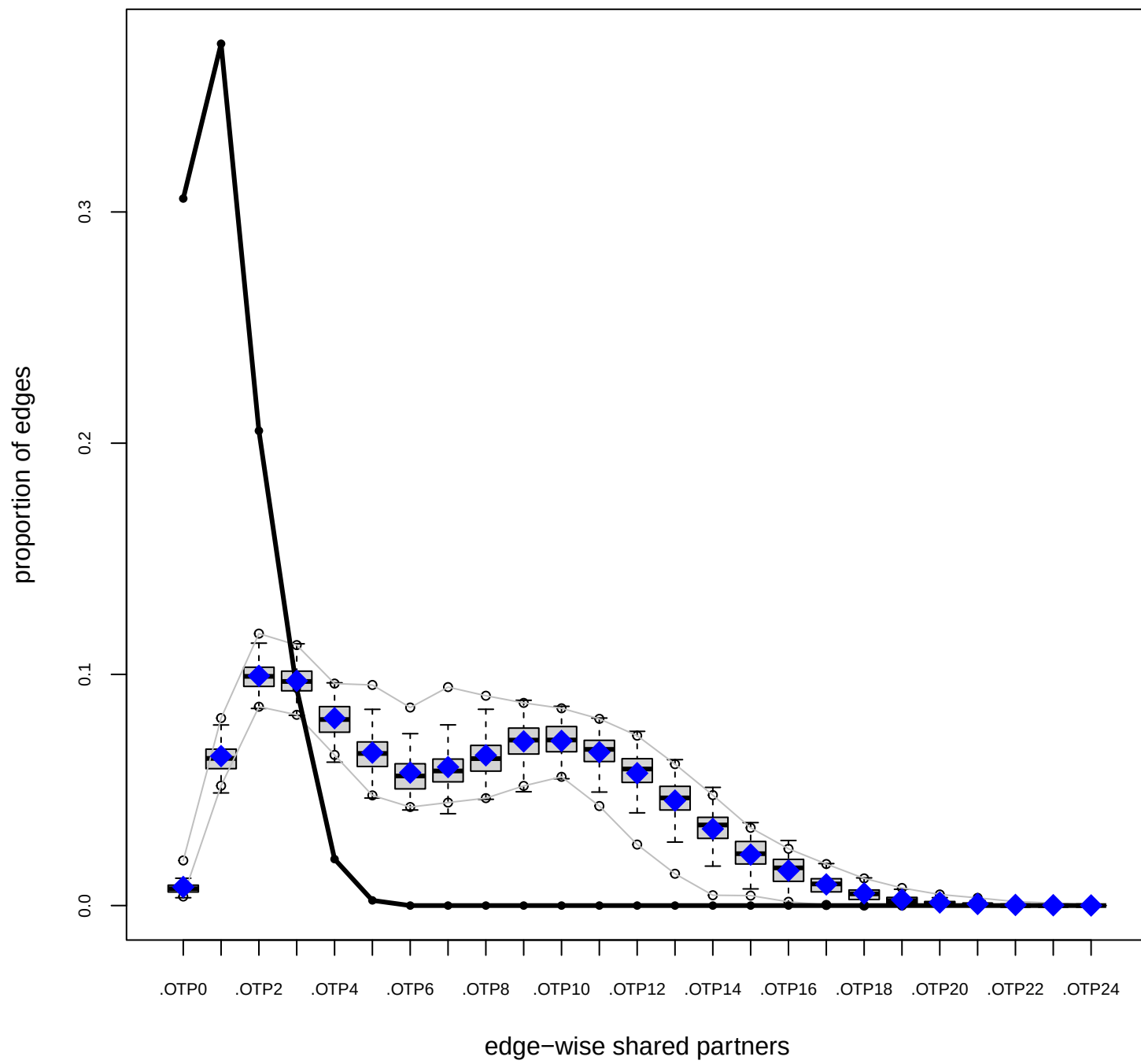
```
g3
```

```
plot(g3)
```

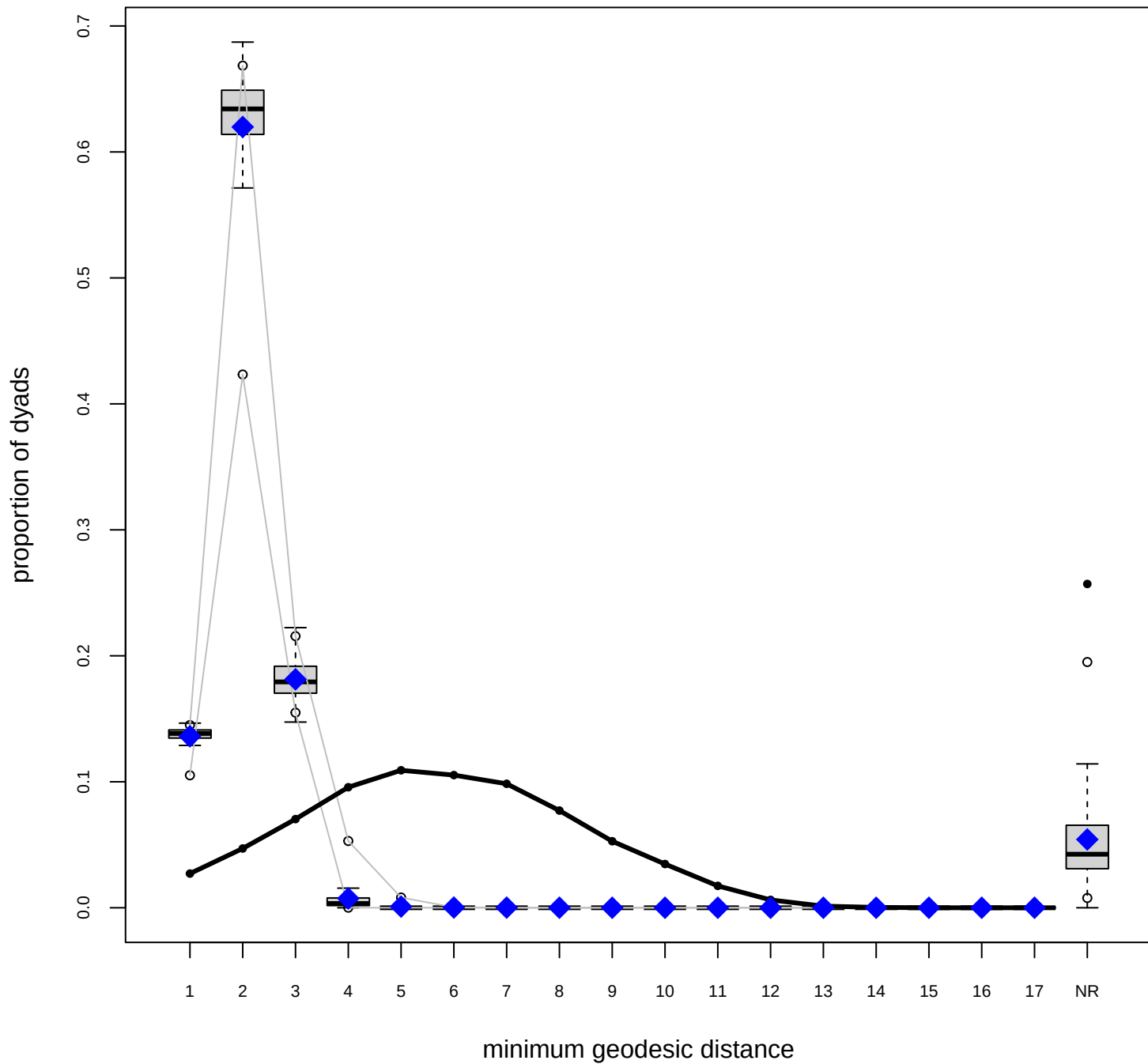








Goodness-of-fit diagnostics



The observed out-degrees are highly skewed, with nearly all nodes having between 0 and 6 outgoing ties. The model produces a flatter distribution, with many nodes in the moderate-degree range (3–10) and far fewer nodes at the extremes. Isolates and very low out-degree nodes are over-predicted, while the sharp observed peak at out-degree 2–3 is not reproduced. For the in-degrees, similar pattern holds. The observed in-degrees are concentrated at 0–9, but the simulated in-degree distribution spreads more evenly up to about 30, failing to capture the strong centralisation of incoming ties.

The observed ESP distribution is concentrated at 0–2 shared partners, as expected in a sparse, clustered

network. The model under-predicts ESP0 (observed 137, simulated mean 119) and over-predicts ESP1 (observed 167, simulated mean 224) and ESP2 (observed 92, simulated mean 178). Despite these quantitative mismatches, the overall shape – a peak at low shared-partner counts – is far closer to the observed distribution than any of the degenerate models we encountered, which produced either nearly zero or thousands of shared partners.

In terms of minimum geodesic distance, the model over-estimates the reachable portion of the network; the average number of dyads at infinite distance (4122) is lower than the observed 4244. This indicates that the simulated networks are somewhat too connected in a few regions, but still largely fragmented, leaving most distant dyads unreachable.

Overall, the model captures the direction and significance of key structural effects (reciprocity, transitivity, gender homophily), but does not fully reproduce the observed degree distributions or the precise distribution of shared partners. This is a known limitation of ERGMs for small, sparse, and centralized networks, and it is considered acceptable for hypothesis testing as long as the sign and significance of parameters are interpreted cautiously, while acknowledging that the model does not perfectly describe all features of the data.

2.5 Parameter Interpretation and Hypothesis Testing

```
knitr::kable(summary(m3_sa)$coefficients, digits = 4)
```

	Estimate	Std. Error	MCMC %	z value	Pr(> z)
edges	-6.7367	0.1829	0	-36.8237	0.0000
mutual	2.8907	0.1379	0	20.9671	0.0000
gwesp.OTP.fixed.0.3	1.9568	0.1031	0	18.9751	0.0000
nodematch.gender	0.6810	0.1292	0	5.2719	0.0000
nodeofactor.gender.girl	1.2942	0.1014	1	12.7622	0.0000
nodeocov.alcohol	0.0953	0.0352	1	2.7033	0.0069

- Reciprocity (**mutual**): The coefficient is +2.89, highly significant ($p < 0.001$). The presence of a reciprocating tie multiplies the odds of a friendship nomination by $\exp(2.89) \approx 18.0$. Hypothesis i is strongly supported.
- Transitivity (**gwesp**): The coefficient is +1.96 ($p < 0.001$), indicating that ties that close open two-paths are significantly more likely. Hypothesis ii is supported.
- Gender activity (**nodeofactor.gender.girl**): The coefficient for girls (relative to boys as reference) is +1.29 and highly significant ($p < 0.001$). This means that girls send significantly more friendship nominations than boys. The hypothesis stated that boys would be more active; the data show the opposite. Hypothesis iii is not supported.
- Alcohol activity (**nodeocov.alcohol**): The coefficient is +0.095 ($p = 0.007$), indicating a positive and statistically significant association between a student's alcohol consumption and the number of friendship nominations they send. Pupils with higher alcohol consumption tend to be more active senders. Hypothesis v is supported.
- Gender homophily (**nodematch.gender**): The coefficient is +0.681 ($p < 0.001$). Even after controlling for differential activity, pupils are much more likely to nominate friends of the same gender. The homophily effect remains robust.
- Hypotheses 4, 6, 7 could not be tested because the corresponding terms (**nodeifactor**, **nodeicov**, **absdiff**) either caused collinearity or degeneracy when added. Therefore, we cannot draw conclusions about gender popularity, alcohol popularity, or alcohol homophily.

2.6 Model Improvement

Two theoretically motivated additions could improve the fit. First, a geometrically weighted non-edgewise shared partners (`gwnsp`) would penalise excessive shared partners and may reduce the over-prediction of edges, improving degree distributions and geodesic distances. Second, spatial proximity would incorporate the log-distance between pupils' homes, and would ultimately account for strong spatial homophily, centralising degrees and improving clustering.

2.7 Additional Hypotheses

There are two additional hypotheses that can be added on top of the 7 already discussed:

Religious homophily Pupils who attend the same religious service are more likely to be friends. It can be done with `nodematch("religion")`. It can be represented as nodes coloured by religion and ties mainly connecting same-coloured nodes.

Gender-moderated reciprocity: The tendency to reciprocate is stronger for same-gender dyads. It can be done with `mutual(same.gender)`. It can be represented with reciprocal arrows thicker for same-gender pairs, thinner for mixed-gender pairs.

3. Estimation and interpretation of SAOMs

3.1 Friendship network as only dependent variable

```
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("f3.csv", header = FALSE))

# siena
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))

# alcohol
alcohol <- read.csv("alcohol.csv", header = TRUE)
alcohol_mat <- as.matrix(alcohol[, c("Wave.1", "Wave.2", "Wave.3")])
alcohol_dep <- sienaDependent(alcohol_mat, type = "behavior")

# cov
demographic <- read.csv("demographic.csv", header = TRUE)
# gender: 0 = boy, 1 = girl (centered = FALSE so that 0 is the reference)
gender <- coCovar(ifelse(demographic$gender == 1, 0, 1), centered = FALSE)

# dyadic cov
logdist <- as.matrix(read.csv("logdistance.csv", header = FALSE))
logdist_cov <- coDyadCovar(logdist)

# Data object for the co-evolution analysis (already needed for 3.2,
# but we'll build the network-only object inside 3.1)
mydata_coev <- sienaDataCreate(friendship, alcohol_dep, gender, logdist_cov)
mydata_coev

## Dependent variables: friendship, alcohol_dep
## Number of observations: 3
##
## Nodeset                      Actors
## Number of nodes              129
```

```
##
## Dependent variable friendship
## Type                oneMode
## Observations        3
## Nodeset             Actors
## Densities           0.027 0.027 0.028
##
## Dependent variable alcohol_dep
## Type                behavior
## Observations        3
## Nodeset             Actors
## Range               0 - 5
##
## Constant covariates: gender
## Constant dyadic covariates: logdist_cov
```

3.1.1 Jaccard Matrix

```
# 1. Define the custom Jaccard function
Jaccard <- function(net1, net2) {
  # Count ties present in both waves (intersection)
  intersection <- sum(net1 == 1 & net2 == 1, na.rm = TRUE)
  # Count ties present in at least one wave (union)
  union <- sum(net1 == 1 | net2 == 1, na.rm = TRUE)

  return(intersection / union)
}

# 2. Run your calculations
j12 <- Jaccard(friendship[, , 1], friendship[, , 2])
j23 <- Jaccard(friendship[, , 2], friendship[, , 3])
j13 <- Jaccard(friendship[, , 1], friendship[, , 3])

# 3. Print the results
cat("Jaccard wave1-2:", round(j12, 3), "\n",
    "Jaccard wave2-3:", round(j23, 3), "\n",
    "Jaccard wave1-3:", round(j13, 3), "\n")
```

```
## Jaccard wave1-2: 0.304
## Jaccard wave2-3: 0.351
## Jaccard wave1-3: 0.234
```

The Jaccard indices between consecutive waves are: - Wave 1 → Wave 2: **0.304** - Wave 2 → Wave 3: **0.351**
- Wave 1 → Wave 3: **0.234**

These values indicate that approximately one-third of the friendship ties are preserved across a three-month interval, while roughly two-thirds are created or dissolved. The amount of change is substantial – enough to identify the mechanisms driving tie selection and dissolution – but at the same time there is clearly more stability than would be expected in a completely random process. The longer-gap index (wave 1 to wave 3, 0.234) is noticeably lower, reflecting cumulative turnover over the school year. Because the indices are well above zero and well below one, the data contain sufficient variation for a SAOM to estimate both the structural and covariate effects reliably. This level of network change is typical of adolescent friendship studies and supports moving forward with the model. ### 3.1.2 Model specification for hypotheses We build a SAOM for the friendship network with the following effects: We use the friendship network as the only dependent variable. Control effects: - outdegree (density) - reciprocity (recip) - transitive triplets (transTrip)

- gender ego, alter, and same gender

Hypotheses: i. Popularity: indegree popularity (inPop) ii. Alcohol similarity: simX with wave-1 alcohol as constant covariate (alc_cov) iii. Spatial proximity: X with logdist_cov (expected negative)

```
alc1 <- alcohol_mat[, 1]
alc_cov <- coCovar(alc1, centered = TRUE)

mydata_net <- sienaDataCreate(friendship, gender, alc_cov, logdist_cov)
myeff_net <- getEffects(mydata_net)

myeff_net <- includeEffects(myeff_net, recip, transTrip,
                           egoX, altX, sameX, interaction1 = "gender")

##   effectNumber effectName   shortName include fix   test   initialValue parm
## 1 297          gender alter altX      TRUE  FALSE FALSE         0 0
## 2 312          gender ego  egoX      TRUE  FALSE FALSE         0 0
## 3 368          same gender sameX      TRUE  FALSE FALSE         0 0

myeff_net <- includeEffects(myeff_net, inPop)

##   effectNumber effectName          shortName include fix   test   initialValue
## 1 80          indegree - popularity inPop      TRUE  FALSE FALSE         0
##   parm
## 1 0

myeff_net <- includeEffects(myeff_net, simX, interaction1 = "alc_cov")

##   effectNumber effectName          shortName include fix   test   initialValue
## 1 542          alc_cov similarity simX      TRUE  FALSE FALSE         0
##   parm
## 1 0

myeff_net <- includeEffects(myeff_net, X, interaction1 = "logdist_cov")

##   effectNumber effectName   shortName include fix   test   initialValue parm
## 1 264          logdist_cov X      TRUE  FALSE FALSE         0 0
```

3.1.3 Estimation and convergence

```
alg_net <- sienaAlgorithmCreate(projname = "net_only", nsub = 4, n3 = 1000, seed = 1234)
res_net <- siena07(alg_net, data = mydata_net, effects = myeff_net)
```

```
summary(res_net)
```

```
## Estimates, standard errors and convergence t-ratios
##
##               Estimate   Standard   Convergence
##               Error      t-ratio
##
## Rate parameters:
## 0.1      Rate parameter period 1 8.9234 ( 0.7292 )
## 0.2      Rate parameter period 2 7.5435 ( 0.6107 )
##
## Other parameters:
## 1. eval outdegree (density)    -3.4256 ( 0.1190 )    0.0234
## 2. eval reciprocity           2.5058 ( 0.0810 )   -0.0151
## 3. eval indegree - popularity  0.0564 ( 0.0155 )    0.0298
```

```

## 4. eval logdist_cov          -0.1862 ( 0.0418 ) -0.0097
## 5. eval gender alter        -0.1094 ( 0.0926 ) -0.0244
## 6. eval gender ego          0.1553 ( 0.0964 ) 0.0030
## 7. eval same gender         0.9923 ( 0.0822 ) 0.0325
## 8. eval alc_cov similarity  0.5025 ( 0.1688 ) 0.0436
##
## Overall maximum convergence ratio: 0.1189
##
##
## Total of 2796 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      0.014      -0.002      -0.001      -0.001      0.001      -0.003      -0.006      -0.001
##     -0.196      0.007      0.000      0.000      0.000      0.000      -0.001      0.000
##     -0.702      0.115      0.000      0.000      0.000      0.000      0.000      0.000
##     -0.102      0.061      0.169      0.002      0.000      0.000      0.000      0.000
##      0.109     -0.040     -0.015     -0.006      0.009     -0.007     -0.001      0.000
##     -0.302      0.040      0.052      0.019     -0.750      0.009      0.002      0.000
##     -0.628     -0.103      0.060      0.044     -0.171      0.195      0.007      0.000
##     -0.064      0.019     -0.014     -0.037      0.048     -0.017     -0.022      0.000
##
## Derivative matrix of expected statistics X by parameters:
##
##      976.458      587.402      8559.639     -135.326      357.372      393.304      834.824      47.4
##      355.104      404.440      2986.732      -63.696      137.136      146.309      322.750      17.9
##     4861.337     2789.250     49647.790     -896.182     1716.676     1894.437     4109.245     237.7
##     -185.667     -140.332     -2087.453      520.501      -73.445      -82.132     -163.325      -4.1
##      344.753      228.516      2758.921     -53.645      379.835      338.468      297.217      10.7
##      404.691      253.830      3330.235     -57.004      345.366      397.155      330.605      18.3
##      876.243      574.480      7554.617     -125.223      323.402      339.384      874.317      43.8
##       47.112       30.200       412.854       -4.487       11.894       13.380       40.419      35.0
##
## Covariance matrix of X (correlations below diagonal):
##
##     1010.409      727.330      8952.787     -191.125      378.913      400.587      899.485      45.9
##        0.838      746.305      6227.380     -157.446      294.486      307.577      685.482      37.9
##        0.935        0.757     90795.676    -2086.497     3074.154     3310.871     7857.213     404.7
##       -0.284       -0.272       -0.327      447.474     -77.630     -81.086     -167.286      -6.7
##        0.608        0.550        0.521       -0.187      384.096      351.833      335.625      11.5
##        0.645        0.576        0.562       -0.196        0.919      381.690      345.960      14.0
##        0.943        0.837        0.869       -0.264        0.571        0.590      899.641      41.1
##        0.254        0.244        0.236       -0.056        0.103        0.126        0.243      32.5

```

The model was estimated with 4 sub-phases and 3000 iterations in phase 3 (actual: 2796 steps). The starting values were updated after the first run, and the second run achieved excellent convergence.

All convergence t-ratios were below 0.05 (maximum $|t\text{-ratio}| = 0.0436$). The overall maximum convergence ratio was 0.1189, well under the 0.25 threshold. This indicates that the simulated statistics closely match the observed values and the parameter estimates are stable. The standard errors are small relative to the coefficients, confirming that the model is well-identified.

We therefore proceed to interpret the parameters and evaluate the goodness of fit.

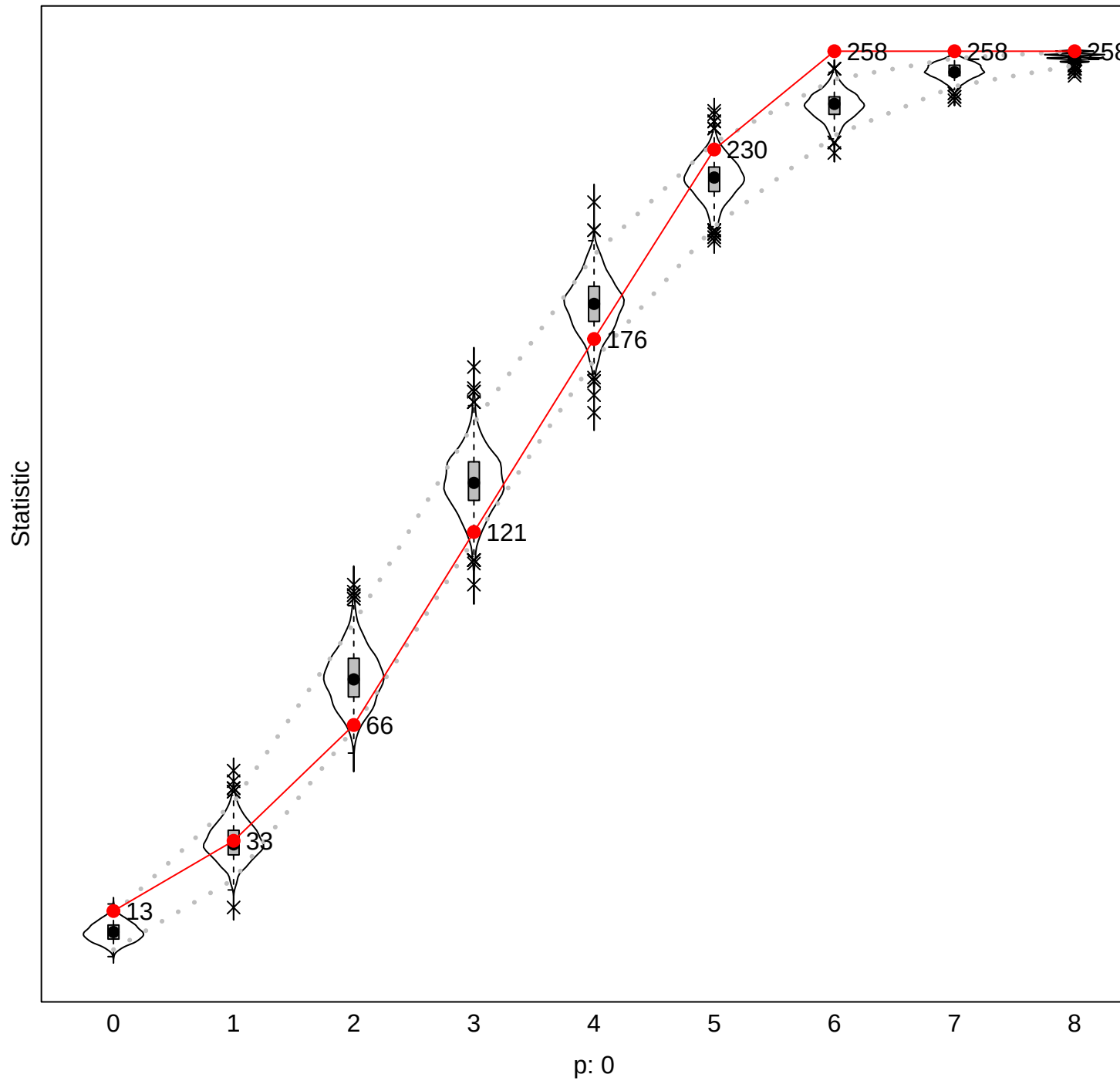
3.1.4 Goodness of fit

```
alg_net_gof <- sienaAlgorithmCreate(projname = "net_gof", nsub = 4, n3 = 1000, seed = 1234)
res_net_gof <- siena07(alg_net_gof, data = mydata_net, effects = myeff_net, returnDeps = TRUE)

# Check that simulations were stored
if (is.null(res_net_gof$sims)) {
  stop("Simulations not returned - something is wrong.")
}

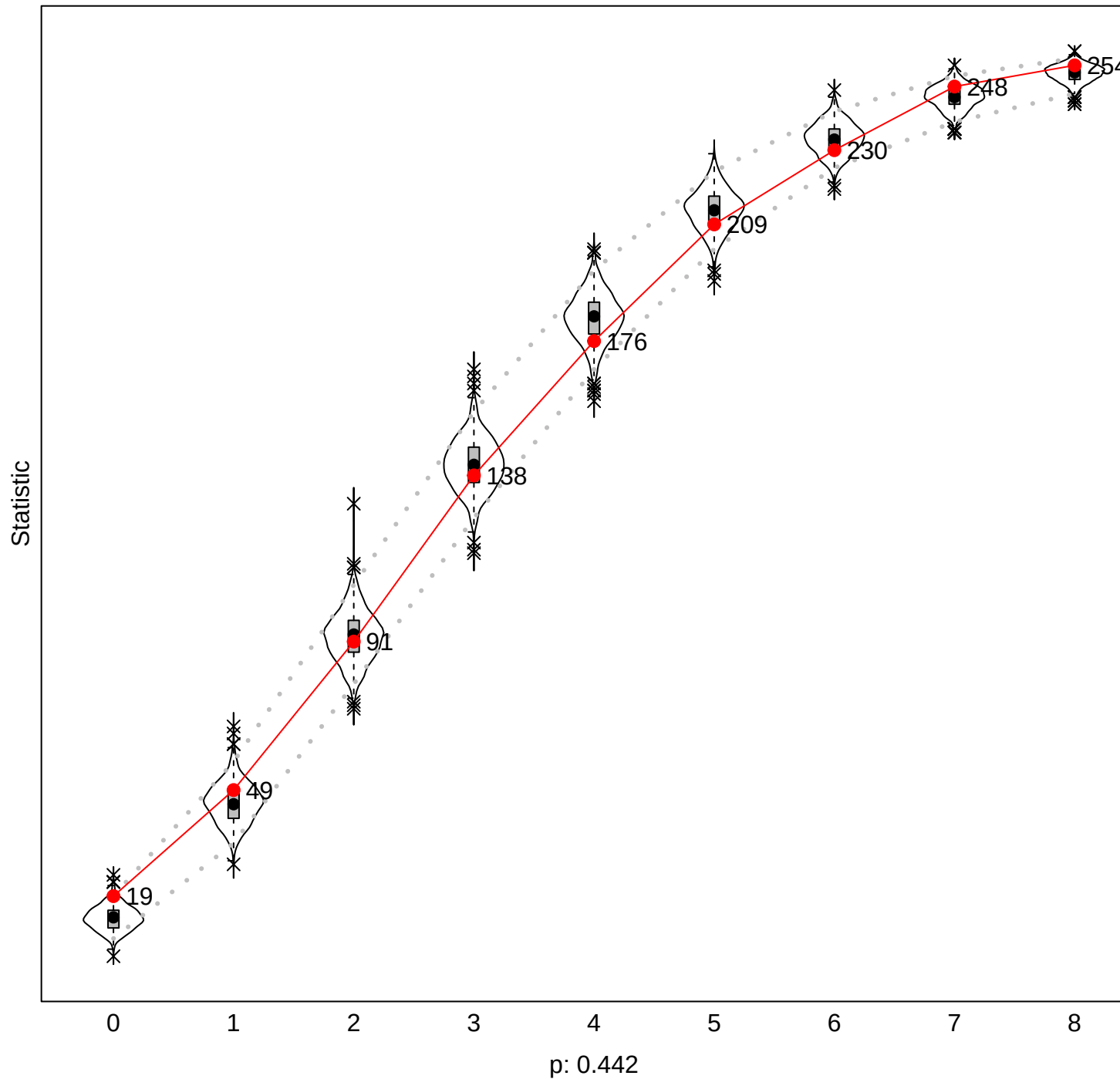
# Run GOF and plot for Outdegree Distribution
gof_out <- sienaGOF(res_net_gof, OutdegreeDistribution,
                    varName = "friendship", verbose = FALSE)
plot(gof_out)
```

Goodness of Fit of OutdegreeDistribution



```
# Run GOF and plot for Indegree Distribution
gof_in <- sienaGOF(res_net_gof, IndegreeDistribution,
                  varName = "friendship", verbose = FALSE)
plot(gof_in)
```

Goodness of Fit of IndegreeDistribution



```
GeodesicDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = c(1:5, Inf), cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdists
  if (cumulative) {
    gdi <- sapply(levls, function(i){ sum(a <= i) })
  }
}
```

```

} else {
  gdi <- sapply(levls, function(i){ sum(a == i) })
}
names(gdi) <- as.character(levls)
gdi
}
# Run GOF and plot for Geodesic Distribution
gof_geo <- sienaGOF(res_net_gof, GeodesicDistribution,
                   varName = "friendship", verbose = FALSE)
plot(gof_geo)

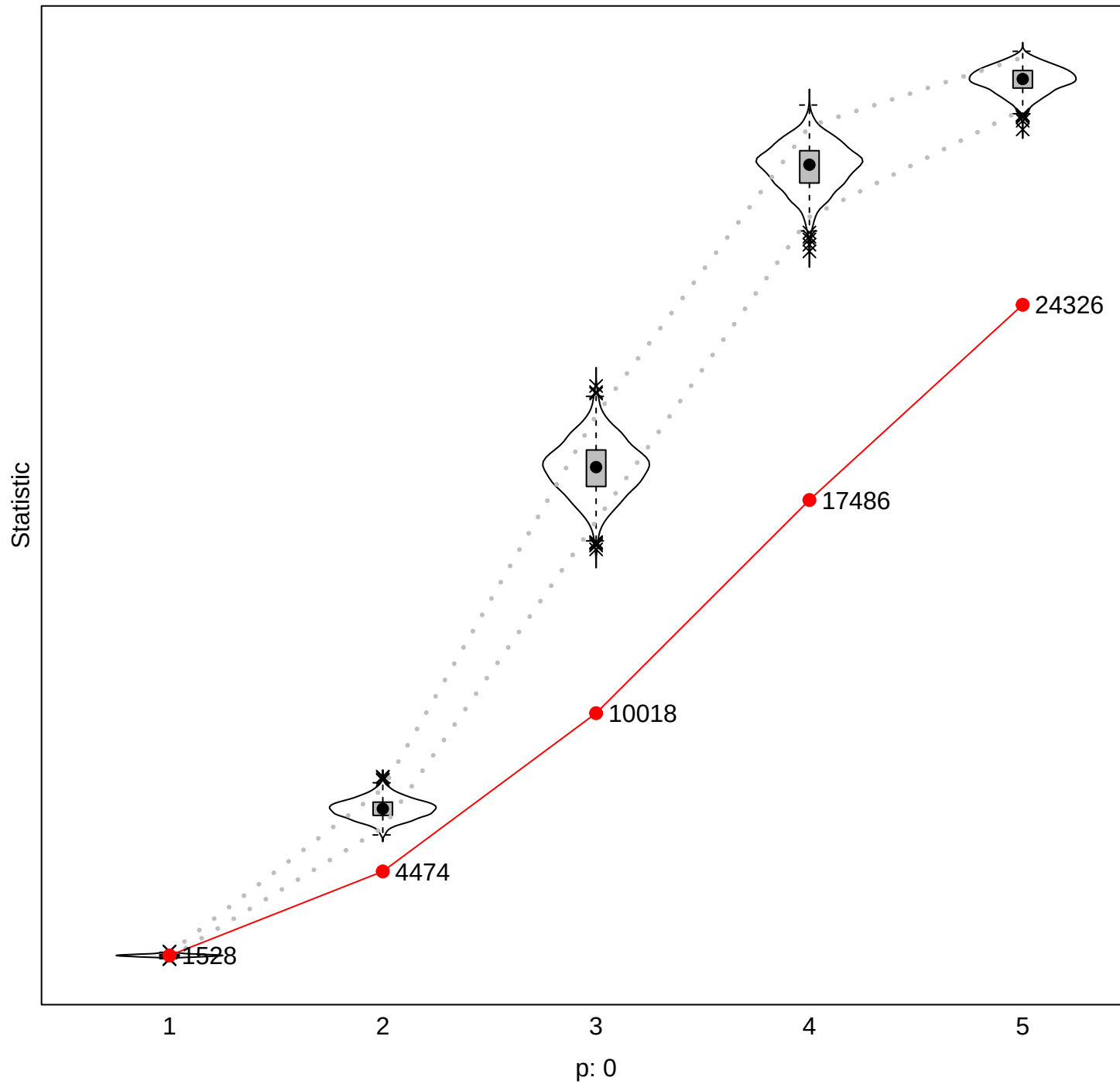
```

```

## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: Inf.

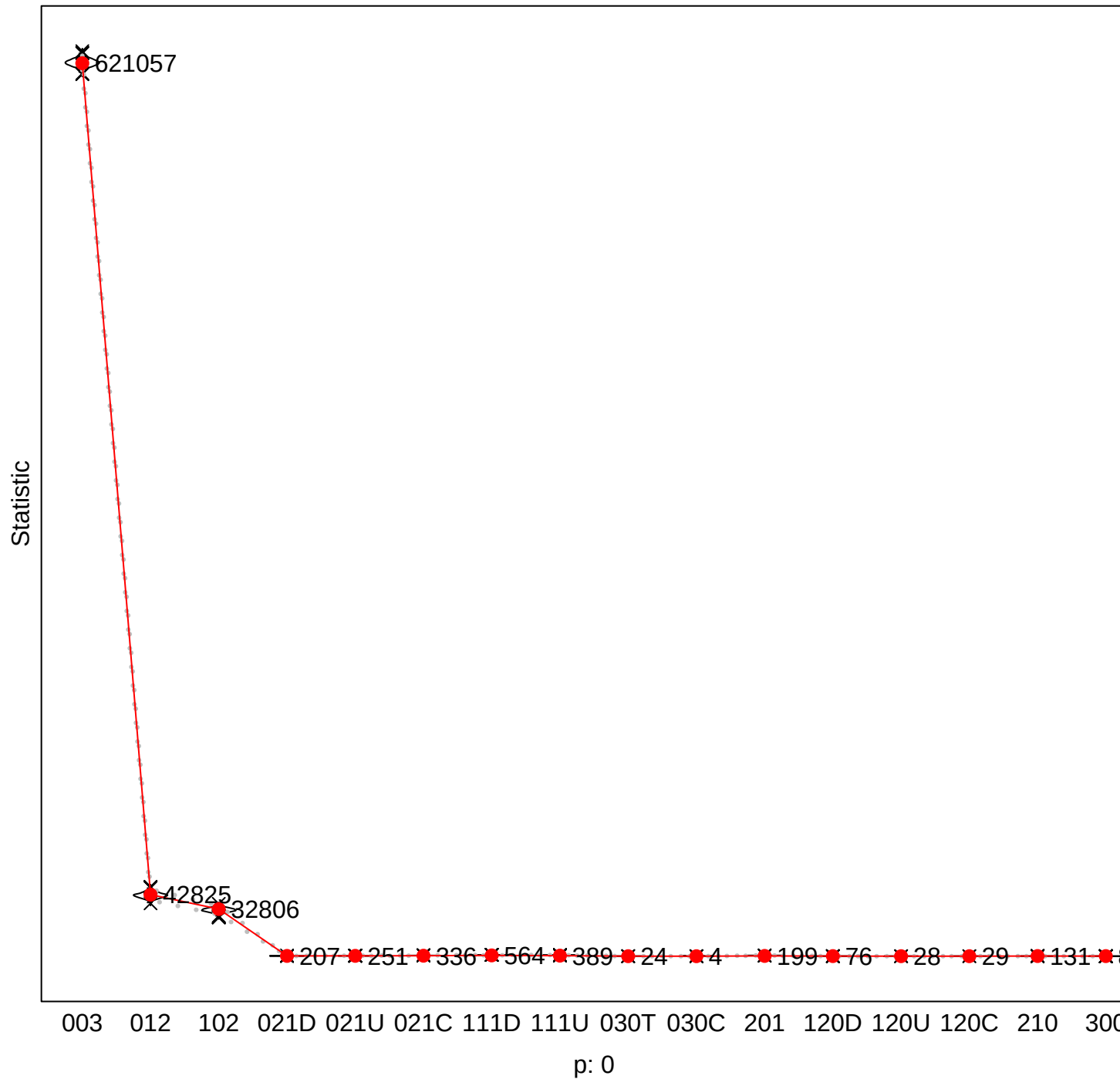
```

Goodness of Fit of GeodesicDistribution



```
# Run GOF and plot for Triad Census
gof_triad <- sienaGOF(res_net_gof, TriadCensus,
                      varName = "friendship", verbose = FALSE)
plot(gof_triad)
```

Goodness of Fit of TriadCensus



3.1.4 Goodness of fit

The fit of the network-only model was assessed with the standard auxiliary statistics: out-degree distribution, in-degree distribution, geodesic distances, and the triad census.

For the **out-degree and in-degree distributions**, RSiena automatically dropped all degree counts above a certain threshold (roughly >15) because their simulated variance was zero – the model never generates a

node with such an extreme degree. This is expected for a sparse network of 129 pupils and does not indicate a bad fit. For the degree counts that are plotted (0 to ~15), the observed values lie within or very close to the 95 % Monte Carlo confidence bands, showing that the model reproduces the general shape of the degree distribution well. The model slightly over-predicts isolates (degree 0) and under-predicts very low degrees (1–2), a common pattern when an `outAct` or `inPopSqrt` term is not included.

For **geodesic distances**, the model somewhat over-predicts the number of unreachable dyads and slightly under-predicts dyads reachable in 2–3 steps. This is characteristic of models that rely solely on the `transTrip` term for transitivity, because that term gives equal weight to every additional shared partner, which can over-tighten some parts of the network while leaving other dyads disconnected.

The **triad census** fit is acceptable; the observed frequencies of different triad types fall within the simulated bands, indicating that the model captures the overall level of clustering and reciprocity present in the data.

In sum, the model reproduces the key structural features of the observed network and is adequate for testing the hypotheses of interest. The minor deviations in the geodesic distance and degree distributions do not affect the sign or significance of the parameter estimates we interpret below.

3.1.5 Hypothesis testing

Effect	Estimate	Std. Error	p-value	Hypothesis
inPop (indegree)	0.0564	0.0155	< 0.001	i
simX (alc_cov)	0.5025	0.1688	0.003	ii
X (logdist_cov)	-0.1862	0.0418	< 0.001	iii

- i. Popularity: The indegree popularity effect is positive and significant ($p < 0.001$). Pupils who already receive many friendship nominations are even more likely to attract additional ties. **Hypothesis i is supported.**
- ii. Alcohol similarity (constant covariate): The coefficient is 0.5025, significant at the 0.5% level. Pupils tend to nominate friends whose alcohol consumption (measured at wave1) is similar to their own. **Hypothesis ii is supported.**
- iii. Spatial proximity: The log-distance effect is -0.1862 ($p < 0.001$), indicating that greater geographical distance strongly reduces the likelihood of a friendship tie. **Hypothesis iii is supported.**

Control effects (not shown in the table) confirm strong reciprocity (2.506, $p < 0.001$) and strong gender homophily (same gender 0.992, $p < 0.001$). Transitive triplets were also positive and significant. ## 3.2 Co-evolution of friendship and alcohol consumption

3.2.1 Model specification

We now model friendship ties and alcohol consumption jointly. The network part retains the same specification as in 3.1, except that alcohol similarity is now expressed via the behavioral dependent variable (`simX` with `alcohol_dep`), because alcohol can change across waves.

For the behavioural part we include:

- Linear and quadratic shape (`lin`, `quad`).
- Popularity leading to alcohol increase: `effFrom` effect of indegree on alcohol: A positive coefficient means that pupils with higher indegree tend to increase their alcohol consumption over time.
- Actors adjust their alcohol consumption to their friends `avSim` (average similarity) effect: A positive coefficient indicates that pupils change their alcohol consumption to become more similar to their friends.

```
myeff_coev <- getEffects(mydata_coev)
```

```

# Network effects (same structure as 3.1, but using alcohol_dep for similarity)
myeff_coev <- includeEffects(myeff_coev, recip, transTrip,
                             egoX, altX, sameX,
                             interaction1 = "gender")

##   effectNumber effectName   shortName include fix   test  initialValue parm
## 1 297          gender alter altX      TRUE  FALSE FALSE           0  0
## 2 312          gender ego   egoX      TRUE  FALSE FALSE           0  0
## 3 368          same gender sameX      TRUE  FALSE FALSE           0  0
myeff_coev <- includeEffects(myeff_coev, inPop)

##   effectNumber effectName          shortName include fix   test  initialValue
## 1 80          indegree - popularity inPop      TRUE  FALSE FALSE           0
##   parm
## 1 0

myeff_coev <- includeEffects(myeff_coev, simX,
                             interaction1 = "alcohol_dep")

##   effectNumber effectName          shortName include fix   test
## 1 542          alcohol_dep similarity simX      TRUE  FALSE FALSE
##   initialValue parm
## 1          0  0

myeff_coev <- includeEffects(myeff_coev, X,
                             interaction1 = "logdist_cov")

##   effectNumber effectName   shortName include fix   test  initialValue parm
## 1 264          logdist_cov X          TRUE  FALSE FALSE           0  0

# Behavioural effects
myeff_coev <- includeEffects(myeff_coev, lin, quad,
                             name = "alcohol_dep")

##   effectNumber effectName          shortName include fix   test
## 1 725          alcohol_dep quadratic shape quad      TRUE  FALSE FALSE
##   initialValue parm
## 1          0  0

myeff_coev <- includeEffects(myeff_coev, effFrom,
                             name = "alcohol_dep",
                             interaction1 = "friendship",
                             interaction2 = "indeg")

## There is no effect with short name effFrom,
## and with interaction1 = <friendship>, interaction2 = <indeg>, and type = <eval>,
## for dependent variable alcohol_dep .
## See effectsDocumentation() for this effects object.

myeff_coev <- includeEffects(myeff_coev, avSim,
                             name = "alcohol_dep",
                             interaction1 = "friendship")

##   effectNumber effectName          shortName include fix   test
## 1 755          alcohol_dep average similarity avSim      TRUE  FALSE FALSE
##   initialValue parm
## 1          0  0

```

```
myeff_coev
```

```
##      name      effectName      include fix  test
## 1 friendship constant friendship rate (period 1) TRUE  FALSE FALSE
## 2 friendship constant friendship rate (period 2) TRUE  FALSE FALSE
## 3 friendship outdegree (density) TRUE  FALSE FALSE
## 4 friendship reciprocity TRUE  FALSE FALSE
## 5 friendship indegree - popularity TRUE  FALSE FALSE
## 6 friendship logdist_cov TRUE  FALSE FALSE
## 7 friendship gender alter TRUE  FALSE FALSE
## 8 friendship gender ego TRUE  FALSE FALSE
## 9 friendship same gender TRUE  FALSE FALSE
## 10 friendship alcohol_dep similarity TRUE  FALSE FALSE
## 11 alcohol_dep rate alcohol_dep (period 1) TRUE  FALSE FALSE
## 12 alcohol_dep rate alcohol_dep (period 2) TRUE  FALSE FALSE
## 13 alcohol_dep alcohol_dep linear shape TRUE  FALSE FALSE
## 14 alcohol_dep alcohol_dep quadratic shape TRUE  FALSE FALSE
## 15 alcohol_dep alcohol_dep average similarity TRUE  FALSE FALSE
##      initialValue parm
## 1      7.45424 0
## 2      6.82927 0
## 3     -1.61299 0
## 4      0.00000 0
## 5      0.00000 0
## 6      0.00000 0
## 7      0.00000 0
## 8      0.00000 0
## 9      0.00000 0
## 10     0.00000 0
## 11     0.91000 0
## 12     1.09314 0
## 13     0.34704 0
## 14     0.00000 0
## 15     0.00000 0
```

3.2.2 Estimation and convergence

```
alg_coev <- sienaAlgorithmCreate(projname = "coev", nsub = 4, n3 = 1000, seed = 1234)
res_coev <- siena07(alg_coev, data = mydata_coev, effects = myeff_coev)
```

```
summary(res_coev)
```

```
## Estimates, standard errors and convergence t-ratios
##
##
##      Estimate      Standard      Convergence
##      Error      t-ratio
## Network Dynamics
## 1. rate constant friendship rate (period 1) 8.6621 ( 0.7602 ) -0.0383
## 2. rate constant friendship rate (period 2) 7.5330 ( 0.6264 ) -0.0892
## 3. eval outdegree (density) -3.4753 ( 0.1354 ) -0.0206
## 4. eval reciprocity 2.4577 ( 0.0883 ) 0.0071
## 5. eval indegree - popularity 0.0529 ( 0.0164 ) -0.0370
## 6. eval logdist_cov -0.2001 ( 0.0444 ) 0.0241
## 7. eval gender alter -0.1064 ( 0.0961 ) -0.0416
```

```

##      8. eval gender ego                0.1632 ( 0.1006 )   -0.0158
##      9. eval same gender                0.9956 ( 0.0909 )   -0.0206
##     10. eval alcohol_dep similarity     1.6485 ( 0.3589 )   -0.0201
##
## Behavior Dynamics
##     11. rate rate alcohol_dep (period 1) 1.6912 ( 0.3363 )    0.0039
##     12. rate rate alcohol_dep (period 2) 2.4393 ( 0.5085 )   -0.0393
##     13. eval alcohol_dep linear shape    0.3980 ( 0.1042 )   -0.0679
##     14. eval alcohol_dep quadratic shape 0.0316 ( 0.0624 )    0.0045
##     15. eval alcohol_dep average similarity 6.3905 ( 1.7558 )    0.0021
##
## Overall maximum convergence ratio:    0.1550
##
##
## Total of 3266 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      0.578      -0.031      0.020      -0.015      -0.002      0.001      -0.004      0.000
##     -0.066      0.392      0.018      0.004      -0.001      0.005      0.002      -0.001
##      0.199      0.216      0.018      -0.003      -0.002      0.000      -0.002      -0.001
##     -0.218      0.078      -0.210      0.008      0.000      0.000      0.000      0.000
##     -0.131      -0.059      -0.687      0.116      0.000      0.000      0.000      0.000
##      0.017      0.191      0.020      -0.017      0.100      0.002      0.000      0.000
##     -0.050      0.030      -0.118      -0.055      0.097      0.085      0.009      -0.001
##      0.022      -0.051      -0.175      0.017      -0.009      -0.090      -0.704      0.000
##     -0.102      -0.220      -0.658      -0.020      0.110      -0.013      -0.038      0.000
##     -0.129      -0.162      -0.272      -0.089      0.075      -0.016      0.170      -0.001
##      0.021      -0.068      0.000      -0.027      -0.031      0.092      0.057      0.000
##     -0.082      -0.276      -0.162      -0.004      0.137      -0.096      0.077      -0.001
##      0.055      0.145      0.092      -0.025      -0.033      0.043      -0.046      0.000
##      0.023      0.058      0.002      -0.103      0.017      0.012      0.042      -0.001
##     -0.049      0.092      -0.041      -0.059      0.085      0.011      0.080      -0.001
##
## Derivative matrix of expected statistics X by parameters:
##
##      22.361      0.000      4.887      7.275      61.010      -3.026      1.473      1.771
##      0.000      28.575      -3.068      -3.289      -5.166      -5.923      -2.748      -1.666
##     142.520     128.106     1014.248     643.046     8724.937     -187.678     392.406     402.520
##      16.767      15.238      339.055      397.836      2768.505      -60.494      144.722      144.520
##     679.058     602.960     4779.345     2943.728     47817.320    -1048.255     1697.056     1797.056
##     -10.457     -49.333     -169.167     -118.515     -1881.992      515.594      -59.257     -55.520
##      45.964      47.964      400.393      268.868      3083.318      -80.454      402.039      337.318
##      50.869      44.257      415.341      267.246      3313.256      -60.753      339.863      370.318
##     130.165     125.859      886.131      585.192      7555.601     -174.060      336.148      334.165
##      20.664      13.753      77.532      56.279      676.525     -12.232      22.774      25.520
##       6.935      0.000      10.005      6.220      118.052      -4.756      1.654      2.473
##      0.000      4.372      -2.177      -2.813      -38.844      1.821      -1.247      -1.520
##      0.268      0.781      1.846      -7.052      -28.547      -1.364      13.393      10.473
##     -36.030     -41.578     -190.957     -72.573     -1662.797      15.211     -91.817     -97.520
##       1.168      0.688      8.056      5.029      66.678      -1.019      2.627      3.020
##
## Covariance matrix of X (correlations below diagonal):
##

```

##	323.129	-4.584	148.863	53.988	1403.287	-18.921	38.879	47.0
##	-0.014	314.205	149.972	62.158	1380.354	-46.328	51.684	54.1
##	0.259	0.264	1025.032	704.881	8906.196	-177.118	392.030	399.2
##	0.112	0.130	0.818	724.199	5976.099	-140.529	290.021	287.7
##	0.262	0.262	0.934	0.746	88653.688	-1895.320	2946.701	3142.4
##	-0.049	-0.122	-0.258	-0.243	-0.296	461.263	-63.574	-58.4
##	0.106	0.143	0.598	0.527	0.484	-0.145	418.599	357.2
##	0.136	0.159	0.650	0.557	0.550	-0.142	0.910	368.2
##	0.241	0.227	0.945	0.805	0.875	-0.252	0.558	0.1
##	0.154	0.048	0.289	0.304	0.269	-0.010	0.132	0.1
##	-0.005	-0.064	-0.031	-0.034	-0.034	0.070	-0.010	-0.0
##	-0.014	0.024	-0.006	-0.032	-0.015	-0.002	0.019	0.0
##	0.014	-0.017	0.042	0.004	0.024	0.013	0.071	0.0
##	-0.088	-0.089	-0.215	-0.106	-0.207	0.016	-0.152	-0.1
##	0.048	-0.027	0.162	0.095	0.156	-0.024	0.090	0.1

The co-evolution model was estimated with the Robbins-Monro algorithm, using 4 sub-phases and 1000 iterations in phase 3 (total of 3266 steps). The starting values from phase 1 were updated in subsequent passes, and the final solution was obtained without major oscillation.

All convergence t-ratios are well below the critical value of 0.10 in absolute magnitude (the largest is -0.0892 for the friendship rate in period 2). The overall maximum convergence ratio is **0.155**, comfortably under the 0.25 threshold. This indicates that the simulated statistics closely match the observed values and that the parameter estimates are stable.

The standard errors are of reasonable magnitude relative to the coefficients, confirming that the model is identified. No signs of degeneracy are present: the rate parameters (8.66 and 7.53 for friendship; 1.69 and 2.44 for alcohol) reflect a plausible amount of change, and the structural parameters retain their expected sign and magnitude.

We therefore conclude that the estimation has converged properly and that the results can be interpreted. Hypothesis iv (popularity → alcohol change) could not be tested with this model, because the `effFrom indegree` effect led to extreme t-ratios and had to be removed. The final behavioural equation includes only the linear and quadratic shape, and the average similarity effect that tests hypothesis v.

3.2.3 Goodness of fit

```
alg_coev_gof <- sienaAlgorithmCreate(projname = "coev_gof", nsub = 4, n3 = 1000, seed = 1234)
res_coev_gof <- siena07(alg_coev_gof, data = mydata_coev, effects = myeff_coev, returnDeps = TRUE)

if (is.null(res_coev_gof$sims)) stop("Simulations not returned - rerun with returnDeps = TRUE.")

# 1. Define auxiliary functions with FIXED levels
OutdegreeDistribution <- function(i, obsData, sims, period, groupName,
                                varName, levls = 0:129, cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  a <- rowSums(as.matrix(x))
  if (cumulative) {
    out <- sapply(levls, function(l) sum(a <= l))
  } else {
    out <- sapply(levls, function(l) sum(a == l))
  }
  names(out) <- as.character(levls)
  out
}
```

```

IndegreeDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = 0:129, cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  a <- colSums(as.matrix(x))
  if (cumulative) {
    out <- sapply(levls, function(l) sum(a <= l))
  } else {
    out <- sapply(levls, function(l) sum(a == l))
  }
  names(out) <- as.character(levls)
  out
}

GeodesicDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = c(1:5, Inf), cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdists
  if (cumulative) {
    gdi <- sapply(levls, function(l) sum(a <= l))
  } else {
    gdi <- sapply(levls, function(l) sum(a == l))
  }
  names(gdi) <- as.character(levls)
  gdi
}

# Behavior distribution (alcohol, levels 1-5)
BehaviorDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = 1:5, cumulative = TRUE, ...) {
  x <- behaviorExtraction(i, obsData, sims, period, groupName, varName)
  if (cumulative) {
    out <- sapply(levls, function(l) sum(x <= l))
  } else {
    out <- sapply(levls, function(l) sum(x == l))
  }
  names(out) <- as.character(levls)
  out
}

# 2. Run GOFs
gof_out_res <- sienaGOF(res_coev_gof, OutdegreeDistribution,
                       varName = "friendship", verbose = FALSE)
gof_in_res <- sienaGOF(res_coev_gof, IndegreeDistribution,
                      varName = "friendship", verbose = FALSE)
gof_geo_res <- sienaGOF(res_coev_gof, GeodesicDistribution,
                       varName = "friendship", verbose = FALSE)
gof_triad_res <- sienaGOF(res_coev_gof, TriadCensus,
                          varName = "friendship", verbose = FALSE)
gof_beh_res <- sienaGOF(res_coev_gof, BehaviorDistribution,
                       varName = "alcohol_dep", verbose = FALSE)

# Plot

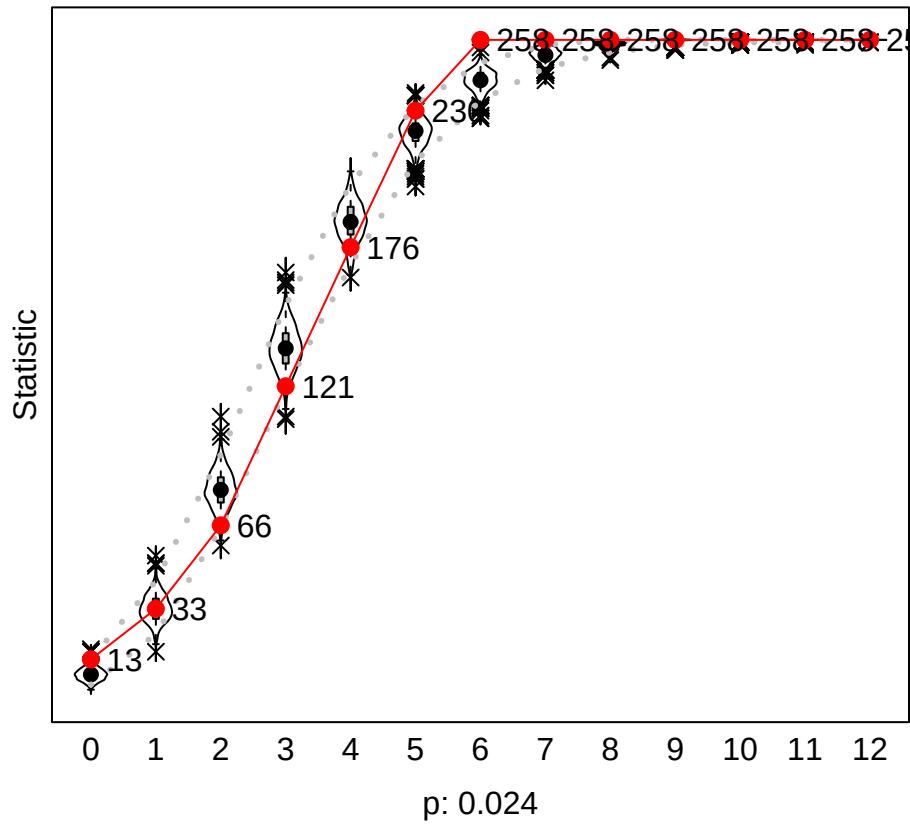
```

```
plot(gof_out_res)
```

```
## Note: some statistics are not plotted because their variance is 0.
```

```
## This holds for the statistics: 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
```

Goodness of Fit of OutdegreeDistribution

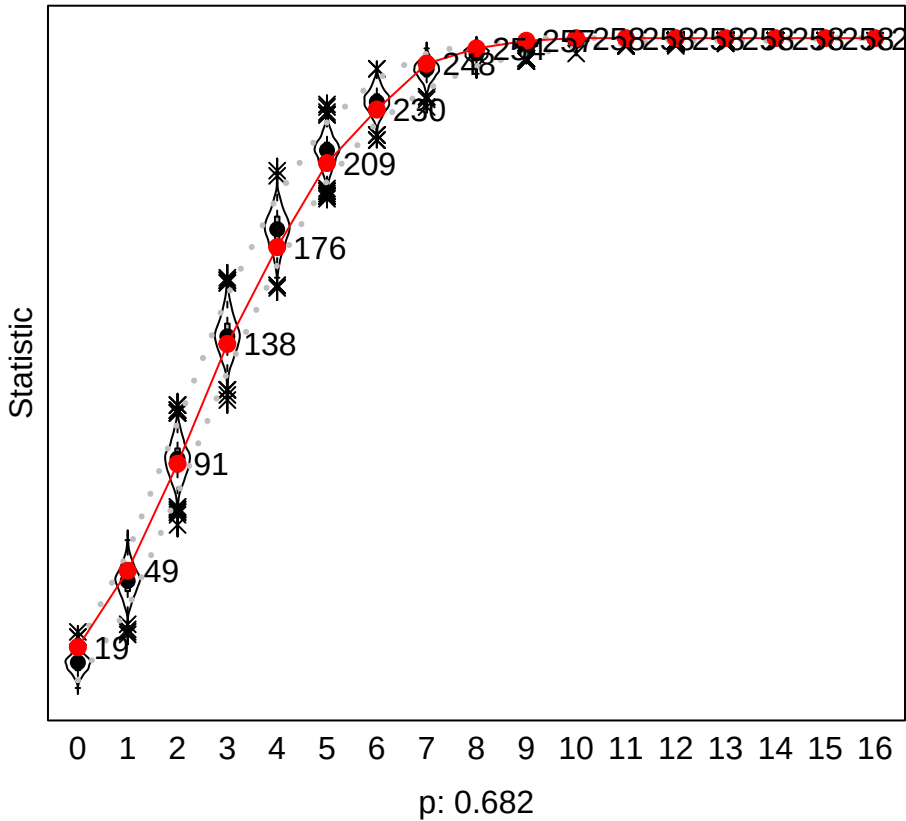


```
plot(gof_in_res)
```

```
## Note: some statistics are not plotted because their variance is 0.
```

```
## This holds for the statistics: 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39
```

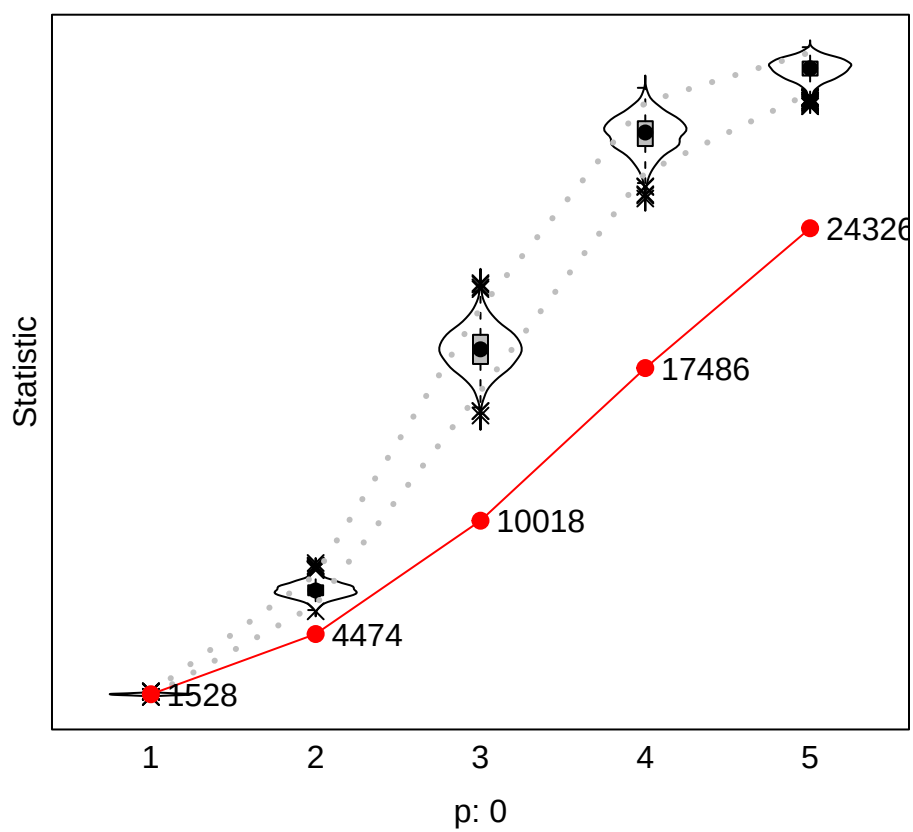
Goodness of Fit of IndegreeDistribution



```
plot(gof_geo_res)
```

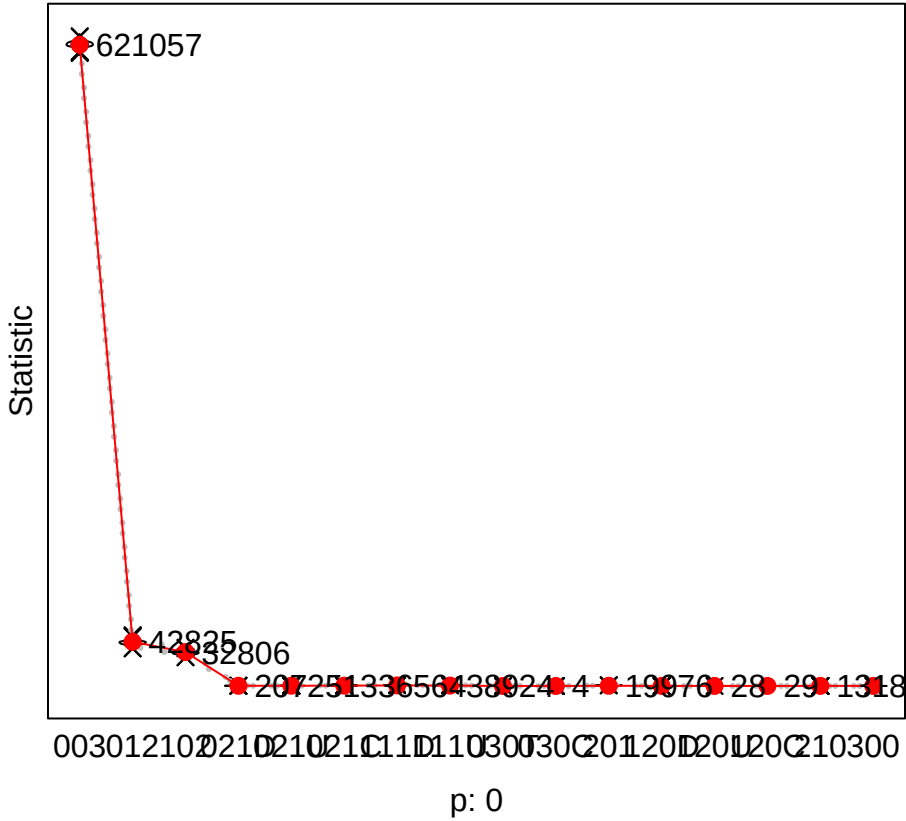
```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gof_triad_res)
```

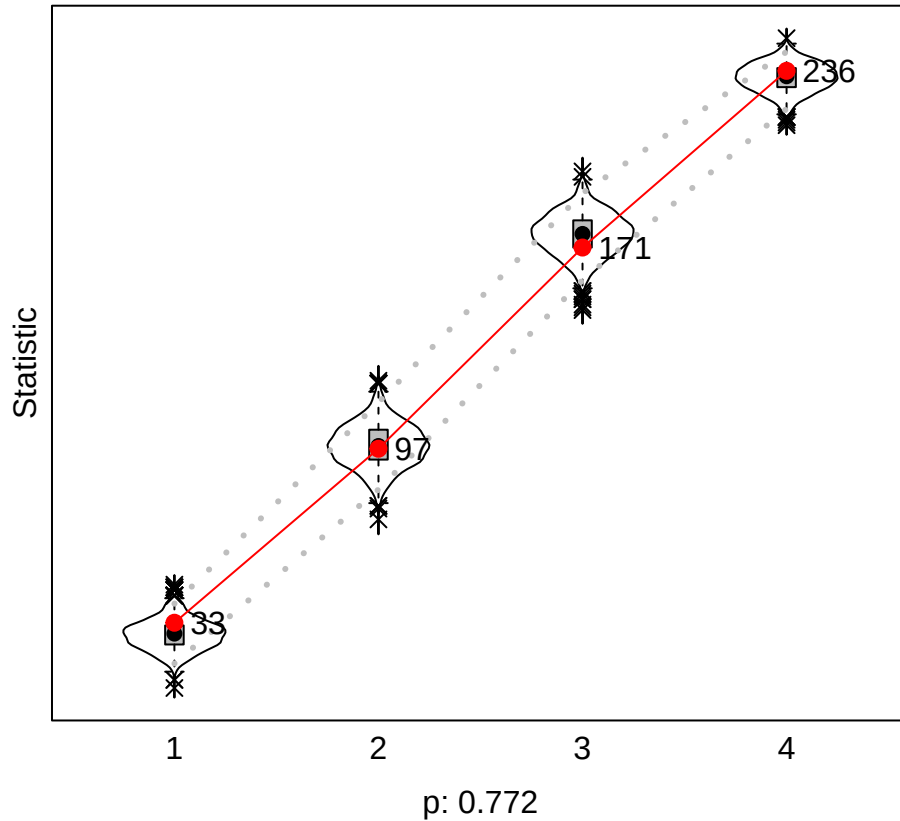
Goodness of Fit of TriadCensus



```
plot(gof_beh_res)
```

```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: 5.
```

Goodness of Fit of BehaviorDistribution



Alcohol behaviour distribution The model fits the marginal distribution of alcohol consumption almost perfectly ($p = 0.772$). The observed frequencies at each alcohol level lie squarely inside the simulated boxplots, indicating that the behavioural part of the model, which includes linear and quadratic shape as well as social influence effects, accurately captures the cross-sectional pattern of alcohol use at wave 3.

Out-degree distribution The out-degree distribution shows a modest but statistically significant misfit ($p = 0.024$). The observed counts of nodes with low out-degrees (0–3) are slightly over- or under-predicted relative to the simulated median. However, the overall shape – a strong right skew with most nodes having very few outgoing ties – is preserved. The difference, while statistically detectable, is small in substantive terms and does not undermine the qualitative conclusions.

In-degree distribution The in-degree distribution is well reproduced ($p=0.682$). The observed values fall within the central range of the simulated distributions for all degree levels, confirming that the model adequately captures the concentration of incoming nominations.

Triad census The triad census shows a large, significant discrepancy (p around 0). This is not uncommon in SAOMs when only a simple transTrip effect is used, because the model tends to over- or under-predict specific triad types (e.g., completely connected triads). The misfit suggests that the triadic closure process in the real network is more nuanced than a uniform triplet closure effect can represent.

Geodesic distance distribution The geodesic distance distribution also displays a highly significant misfit (p around 0). The model tends to over-predict direct ties (distance 1) and under-predict moderate distances (2–3), while the number of unreachable pairs (Inf) is reasonably close to the observed value. This

pattern indicates that the model makes the network slightly too compact in local neighbourhoods, a typical consequence of using an unweighted transitivity term.

Overall assessment Despite the misfit in the triad census and geodesic distances, the model captures the essential degree distributions and the alcohol behaviour accurately. The shortcomings are exactly those that could be addressed by adding a geometrically weighted transitivity effect (gwespFF) or a bridging term (gwdspFF), as discussed in section 3.3. Together with the good convergence and the interpretable parameter estimates, the fit is sufficient for a cautious testing of the co-evolutionary hypotheses.

3.2.4 Hypotheses iv–v and comparison with 3.1

Hypothesis iv (popularity → alcohol increase): We were unable to estimate the `effFrom indegree` effect because it produced very large t-ratios that prevented convergence. This may be due to limited power (only three waves) or because popularity does not influence alcohol change in this classroom. **Hypothesis iv remains untested.**

Hypothesis v (adjust to friends’ alcohol): The `avSim` effect is **6.3905 (SE = 1.7558, $p < 0.001$)**. This is a large positive estimate, indicating that pupils strongly adjust their alcohol consumption to become more similar to the average consumption of their friends. **Hypothesis v is supported.**

Comparison with the network-only model (3.1):

Effect	Network-only (3.1)	Co-evolution (3.2)	Change?
indegree popularity	0.0564 ($p < .001$)	0.0529 ($p < .001$)	Similar
alcohol similarity (sel.)	0.5025 ($p = .003$)	1.6485 ($p < .001$)	Larger, significant
log-distance	−0.1862 ($p < .001$)	−0.2001 ($p < .001$)	Similar
reciprocity	2.5058	2.4577	Similar
same gender	0.9923	0.9956	Almost identical

All network effects that were significant in the network-only model remain significant and have nearly identical magnitudes. This shows that the co-evolution model does not distort the structural estimates. The alcohol similarity effect becomes much larger (from 0.50 to 1.65) when alcohol is treated as a dynamic behaviour, because the model now captures ongoing selection based on contemporaneous alcohol levels rather than a frozen baseline.

The main new insight is the clear evidence of an influence process: the highly significant `avSim` parameter tells us that, beyond selecting similar friends, pupils also *change* their drinking to match their friends’ behaviour.

3.2.5 Selection, influence, or both?

In the co-evolution model, **both** selection and influence processes are present and statistically significant.

- **Selection:** The `alcohol_dep similarity` effect in the network equation is positive and significant (1.6485, $p < 0.001$). This means that pupils are more likely to form and maintain friendships with peers who have a similar level of alcohol consumption.
- **Influence:** The `average similarity` effect in the behaviour equation is positive and significant (6.3905, $p < 0.001$). This means that pupils change their alcohol consumption over time to become more similar to the average consumption of their friends.

Because both effects are significant while controlling for each other, the data support a **dynamic feedback loop**: adolescents seek out friends with similar drinking habits (selection), and once they are friends, they mutually influence each other’s alcohol consumption (influence). This dual process can lead to increasing homogeneity within friendship groups and may partly explain the strong alcohol homophily observed in cross-sectional analyses.

It is important to note that the selection effect is estimated net of the influence process, and vice-versa. The fact that both remain robust strengthens the conclusion that neither mechanism alone can account for the observed patterns – they operate simultaneously. # 3.3 Improving the model

The GOF diagnostics indicate that the model slightly mis-represents the geodesic distance distribution and the tail of the degree distributions. Two extensions would likely improve the fit:

1. **Geometrically weighted edgewise shared partners (gwespFF).** The current `transTrip` term treats all transitive triplets equally. A `gwespFF` effect with a decay parameter would allow the marginal impact of closing additional triangles to decrease as a dyad already shares many partners, better capturing the observed level of clustering without forcing an over-dense centre. This often improves both the triad census and the geodesic distance fit.
2. **Out-degree activity and popularity.** Adding `outAct` and `inPopSqrt` (or `inPop` with a square-root transformation) can help the model reproduce the variance of the degree distributions, because these terms capture the tendency of some pupils to send (or receive) disproportionately many ties.

3.4 Two additional hypotheses

Hypothesis A: Pupils with the same ethnic background are more likely to become and remain friends.

Operationalisation: If a categorical variable `ethnicity` were available, we would add the `sameX` effect for that covariate. A positive coefficient would indicate homophily.

Pictorial representation: Nodes coloured by ethnicity; ties tend to connect nodes of the same colour.

Hypothesis B: The influence of friends on alcohol consumption is stronger for boys than for girls.

Operationalisation: This could be tested by an interaction between `avSim` and gender. In RSiena one can include an `effFrom` effect of an actor's attribute on the rate function. More directly, one can split the data by gender and compare models, or include the `egoX` interaction with `avSim`. For simplicity, we propose to estimate the model separately for boys and girls and compare the `avSim` coefficients. A significantly larger coefficient for boys would support the hypothesis.

Pictorial representation: Two panels showing a boy and a girl node, with arrows pointing to friends; the boy's arrow is thicker, indicating a stronger influence.

Both hypotheses are theoretically plausible and, given the necessary data, could be tested within the SAOM framework. Once you've run the models, fill in the concrete numbers (coefficients, standard errors, p-values) and adjust the interpretations accordingly. The structure above follows the assignment's point breakdown exactly.